



The Dentistry —INSIGHT—

AUSTRALIAN DENTAL THERAPEUTIC GUIDELINE (TG)

A Simple, Picture-Based Clinical Reference Compilation

*Compiled from The Dentistry Insight's educational clinical notes
and the Australian Dental Therapeutic Guidelines*



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This document is a simple, picture-based clinical reference compiled for quick revision of dental therapeutic and clinical guidelines, prepared and formatted by The Dentistry Insight for educational purposes.

Credit / Source Acknowledgement:

The therapeutic and drug-of-choice information referenced in this guideline is based on the Australian Dental Therapeutic Guidelines. Full credit for the original clinical guidance goes to the Australian Dental Therapeutic Guidelines.

For complete, up-to-date and official clinical guidance, please always refer to the official Australian Dental Association (ADA) website.

This compilation has been prepared twice / re-issued by the author for internal educational use, with all diagrams and clinical notes credited to their original source.

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Part B - Clinical Picture Gallery (Australian Dental Therapeutic Guidelines)

Illustrated clinical reference images, presented simply, one per page.

Major Connector

- **Lingual Bar:** Provides support and stability; suitable where sufficient space exists between the tongue and alveolar ridge.
- **Lingual Plate:** Wider, flat plate providing additional support and stability.
- **Sublingual Bar:** Suitable for limited space; follows the contour of the floor of the mouth.
- **Cingulum Bar (Continuous Clasp):** Rests on the cingulum of anterior teeth; provides additional support.

Minor Connector

Connects the major connector to the other components of the framework (rests, clasps, bases).

Rest

- **Occlusal Rest:** Provides vertical support on occlusal surfaces of posterior teeth.
- **Cingulum Rest:** Provides support on the cingulum of anterior teeth.

Clasp Assembly

- **Circumferential Clasp:** Provides retention by encircling the tooth.
- **Bar Clasp:** Provides retention via a bar-shaped design approaching from a gingival direction.

Base (Saddle)

Covers the edentulous area; provides support, stability and carries the artificial teeth.

Artificial Teeth

Replace the missing natural teeth and restore function and aesthetics.

Functions of an RPD

- Support
- Stability
- Retention

Prosthodontics High-Yield Points

Alveolar Ridge Restoration & Atwood Classification

Before fabricating complete dentures, assessing the alveolar ridge is crucial. Alveolar ridge restoration helps restore the ridge's shape and function. Understanding the Atwood classification system, the pattern of ridge resorption, and the consequences of ridge resorption can guide treatment planning.

Atwood Classification

- 1 **Class I:** Pre-extraction ridge form; records available, ideal for restoration.
- 2 **Class II:** High, well-rounded ridge; minimal resorption.
- 3 **Class III:** Knife-edge ridge; moderate resorption.
- 4 **Class IV:** Flat ridge; significant resorption.
- 5 **Class V:** Depressed ridge; advanced resorption.
- 6 **Class VI:** Minimal ridge; extreme resorption.

Pattern of Ridge Resorption

- 1 **Mandible:** Resorption occurs more rapidly and is typically more pronounced in the labial and lingual direction, resulting in widening of the posterior arch.
- 2 **Maxilla:** Resorption is generally slower and more uniform, and resorbs buccally and labially.

Consequences of Ridge Resorption

- 1 **Denture instability:** Reduced retention and support.
- 2 **Aesthetic concerns:** Changes in facial structure.
- 3 **Functional issues:** Difficulty chewing and speaking.
- 4 **Bone loss:** Further resorption can occur.

Importance of Ridge Restoration

- 1 **Improved denture stability:** Enhanced retention and support.
- 2 **Better aesthetics:** Restored facial contours.
- 3 **Enhanced function:** Improved chewing and speaking.

1. Toxicity Thresholds

- Safely Tolerated Dose (STD): 1 mg/kg body weight
- Potentially Lethal Dose (PLD): 5 mg/kg body weight
- Certainly Lethal Dose (CLD): 32–64 mg/kg body weight

2. Fluoride Concentrations in Products

- Standard fluoride toothpaste: 1000 ppm F ($\approx$ 1 mg F/mL)
- Daily fluoride mouthrinse: 0.23 mg F/mL
- APF gel: 12.3 mg F/mL
- Fluoride varnish: 22.6 mg F/mL

3. Ingestion Amounts for Toxicity

A 19 kg child would need to ingest a significant amount of a fluoride product to reach a toxic level — e.g. approximately 63 mL of 1500 ppm toothpaste, or 7.6 mL of 1.23% APF gel, to reach the hospitalisation threshold.

4. Antidotes and Treatment

- <5 mg F/kg body weight: give a large volume of milk.
- >5 mg F/kg body weight: refer to hospital for gastric lavage; consider IV calcium gluconate and an emetic if presentation is delayed.

Clinical Evaluation of Tooth Wear

Attrition | Erosion | Abfraction | Abrasion

Clinical Evaluation Protocols

The Lateral Excursion Test (Occlusal Analysis) – Have the patient bite down into Maximum Intercuspation Position (MIP) and perform lateral (side-to-side) grinding movements.

Abfraction indicator: heavy, non-axial contact or interference is observed on the posterior teeth during these movements. If the canine tip is worn down, it forces the posterior teeth to absorb the entire lateral load – causing the tooth neck to flex and enamel rods to break away.

Abrasion indicator: the posterior teeth glide smoothly with no unexpected interference, suggesting the wear is due to external mechanical factors such as aggressive tooth-brushing.

Location and Mapping Pattern

Abrasion patterns typically present as a continuous, “swept” pattern spanning several teeth at the cervical margin.

(Note: the source material for this section is a partially truncated message — further detail on erosion and attrition mapping was not captured in the original note.)

- 1 Sympathetic stimulation increases heart rate.
- 2 Parasympathetic stimulation decreases heart rate.
- 3 Beta-1 receptors are mainly present in the heart.
- 4 Alpha-1 receptors cause vasoconstriction.
- 5 Beta-2 receptors cause bronchodilation.
- 6 Norepinephrine mainly acts on alpha receptors.
- 7 Epinephrine acts on alpha and beta receptors.
- 8 Dopamine increases renal blood flow at low dose.
- 9 Dobutamine mainly stimulates beta-1 receptors.
- 10 Phenylephrine is a pure alpha agonist.
- 11 Isoproterenol is a nonselective beta agonist.
- 12 Clonidine is an alpha-2 agonist.
- 13 Alpha blockers cause postural hypotension.
- 14 Prazosin selectively blocks alpha-1 receptors.
- 15 Propranolol is a nonselective beta blocker.
- 16 Atenolol is a beta-1 selective blocker.
- 17 Beta blockers decrease heart rate.
- 18 Beta blockers decrease renin release.
- 19 Beta blockers are contraindicated in asthma.
- 20 Abrupt withdrawal of beta blockers causes rebound hypertension.

Antihypertensive Drugs

High-Yield Points (continued)

- 21 Hypertension is the “silent killer.”
- 22 ACE inhibitors are first-line antihypertensives.
- 23 ACE inhibitors decrease angiotensin II.
- 24 ACE inhibitors increase bradykinin.
- 25 ACE inhibitors cause dry cough.
- 26 ACE inhibitors may cause hyperkalemia.
- 27 ACE inhibitors are contraindicated in pregnancy.
- 28 Enalapril is an ACE inhibitor.
- 29 Losartan is an ARB.
- 30 ARBs do not cause cough.
- 31 Aliskiren is a direct renin inhibitor.
- 32 Calcium channel blockers cause vasodilation.
- 33 Amlodipine is a dihydropyridine CCB.
- 34 Verapamil mainly affects the heart.
- 35 Diltiazem decreases AV conduction.
- 36 Verapamil is contraindicated in heart failure.
- 37 Nifedipine may cause reflex tachycardia.
- 38 Thiazides are first-line in hypertension.
- 39 Hydrochlorothiazide may cause hypokalemia.
- 40 Thiazides increase calcium reabsorption.
- 41 Loop diuretics act on the thick ascending limb.
- 42 Furosemide causes hypokalemia.
- 43 Furosemide causes ototoxicity.
- 44 Ethacrynic acid is a non-sulfa loop diuretic.
- 45 Spironolactone is a potassium-sparing diuretic.
- 46 Spironolactone antagonises aldosterone.
- 47 Spironolactone may cause gynecomastia.
- 48 Eplerenone causes less gynecomastia.

Antianginal & Antiarrhythmic Drugs

High-Yield Points

- 49 Mannitol is an osmotic diuretic.
- 50 Acetazolamide inhibits carbonic anhydrase.
- 51 Hydralazine causes a lupus-like syndrome.
- 52 Minoxidil causes hypertrichosis.
- 53 Nitroprusside is used in hypertensive emergency.
- 54 Methyldopa is safe in pregnancy.
- 55 Labetalol is used in hypertensive emergency.

Antianginal Drugs

- 56 Angina occurs due to myocardial ischemia.
- 57 Nitrates are vasodilators.
- 58 Nitroglycerin decreases preload.
- 59 Nitrates increase cGMP.
- 60 Headache is a common side effect of nitrates.
- 61 Nitrates are contraindicated with sildenafil.
- 62 Sildenafil inhibits PDE-5.
- 63 Beta blockers decrease myocardial oxygen demand.
- 64 Calcium channel blockers relieve vasospasm.
- 65 Verapamil is useful in Prinzmetal angina.
- 66 Ranolazine inhibits the late sodium current.
- 67 Stable angina is treated with nitrates.
- 68 Unstable angina may progress to MI.
- 69 Aspirin reduces the risk of MI.
- 70 Statins reduce LDL cholesterol.

Antiarrhythmic Drugs

- 71 Class I drugs block sodium channels.
- 72 Quinidine is a class IA drug.

- 71 Class I drugs block sodium channels.
- 72 Quinidine is a class IA drug.
- 73 Procainamide may cause a lupus-like syndrome.
- 74 Lidocaine is a class IB drug.
- 75 Lidocaine is useful in ventricular arrhythmias.
- 76 Flecainide is a class IC drug.
- 77 Beta blockers are class II drugs.
- 78 Amiodarone is a class III drug.
- 79 Amiodarone prolongs action potential duration.
- 80 Amiodarone causes pulmonary fibrosis.
- 81 Amiodarone causes thyroid dysfunction.
- 82 Sotalol has beta-blocking action.
- 83 Verapamil is a class IV antiarrhythmic.
- 84 Adenosine is used in PSVT.
- 85 Adenosine has a very short half-life.
- 86 Digoxin increases vagal tone.
- 87 Magnesium is useful in torsades de pointes.
- 88 Quinidine may cause cinchonism.
- 89 Torsades de pointes is associated with prolonged QT.
- 90 Amiodarone is a broad-spectrum antiarrhythmic.

Heart Failure Drugs

- 91 Digoxin increases cardiac contractility.
- 92 Digoxin inhibits Na/K ATPase.
- 93 Digoxin toxicity causes arrhythmias.
- 94 Yellow vision is seen in digoxin toxicity.

Heart Failure Drugs & Anticoagulants

Antiplatelet Drugs & High-Yield Points

- 91 Digoxin increases cardiac contractility.
- 92 Digoxin inhibits Na/K ATPase.
- 93 Digoxin toxicity causes arrhythmias.
- 94 Yellow vision is seen in digoxin toxicity.
- 95 Hypokalemia predisposes to digoxin toxicity.
- 96 BNP is elevated in heart failure.
- 97 ACE inhibitors improve survival in CHF.
- 98 Beta blockers are beneficial in chronic CHF.
- 99 Diuretics reduce oedema in CHF.
- 100 Dobutamine is used in acute heart failure.
- 101 Milrinone is a PDE-3 inhibitor.
- 102 Nesiritide is recombinant BNP.
- 103 Sacubitril inhibits neprilysin.
- 104 ARNI = sacubitril + valsartan.
- 105 Chronic CHF activates the RAAS.

Anticoagulants & Antiplatelets

- 106 Aspirin irreversibly inhibits COX.
- 107 Aspirin decreases TXA2 synthesis.
- 108 Clopidogrel blocks ADP receptors.
- 109 Ticagrelor reversibly blocks the P2Y12 receptor.
- 110 GP IIb/IIIa blockers inhibit platelet aggregation.
- 111 Abciximab is a GP IIb/IIIa blocker.
- 112 Heparin activates antithrombin III.
- 113 Heparin is monitored by aPTT.
- 114 Protamine sulfate reverses heparin.
- 115 LMWH has a more predictable effect.
- 116 Enoxaparin is an LMWH.
- 117 Warfarin inhibits vitamin K epoxide reductase.

Shock & Emergency Drugs

CVS Drug Side Effects

- 146 Norepinephrine is used in septic shock.
- 147 Dopamine is used in cardiogenic shock.
- 148 Atropine is used in bradycardia.
- 149 Adenosine terminates PSVT.
- 150 Amiodarone is used in ventricular tachycardia.
- 151 Magnesium is used in torsades de pointes.
- 152 Epinephrine is used in anaphylactic shock.
- 153 Dobutamine increases cardiac output.
- 154 Vasopressin causes vasoconstriction.
- 155 Atropine blocks muscarinic receptors.

CVS Drug Side Effects

- 156 ACE inhibitors cause cough.
- 157 ACE inhibitors cause angioedema.
- 158 ARBs cause hyperkalemia.
- 159 Beta blockers cause bradycardia.
- 160 Verapamil causes constipation.
- 161 Nifedipine causes oedema.
- 162 Minoxidil causes hypertrichosis.
- 163 Thiazides cause hyperglycemia.
- 164 Loop diuretics cause hypokalemia.
- 165 Digoxin causes xanthopsia.
- 166 Statins cause rhabdomyolysis.
- 167 Warfarin causes skin necrosis.
- 168 Heparin causes osteoporosis with long-term use.
- 169 Hydralazine causes reflex tachycardia.
- 170 Spironolactone causes gynecomastia.

Most Repeated Exam Pearls

Cardiovascular Pharmacology

- 171 Most common side effect of ACE inhibitors — cough.
- 172 Drug causing lupus-like syndrome — hydralazine.
- 173 Drug causing gingival hyperplasia — nifedipine.
- 174 Drug causing pulmonary fibrosis — amiodarone.
- 175 Drug causing ototoxicity — furosemide.
- 176 Drug safest in pregnancy HTN — methyldopa.
- 177 Drug causing gynecomastia — spironolactone.
- 178 Drug used in PSVT — adenosine.
- 179 Drug used in torsades — magnesium sulfate.
- 180 Drug contraindicated with nitrates — sildenafil.
- 181 Most specific MI marker — troponin I.
- 182 Best drug after MI — beta blockers.
- 183 Aspirin decreases platelet aggregation.
- 184 Clopidogrel blocks the ADP receptor.
- 185 Heparin works immediately.
- 186 Warfarin has a delayed onset.
- 187 INR monitors warfarin.
- 188 aPTT monitors heparin.
- 189 Protamine reverses heparin.
- 190 Vitamin K reverses warfarin.

Vertical Root Fractures

12 Diagnostic Points You Will Remember

- 1 Pain – mild or spontaneous on mastication.
- 2 Sinus tract – single or multiple.
- 3 Isolated pockets – extending from the gingival sulcus to the apex.
- 4 Abscess – periodontal type.
- 5 Fracture line – complete or incomplete.
- 6 Radiographic sign – a “J-shaped” radiolucency, a radiolucent halo surrounding the root surface.

Recommended treatment: **Extraction.**

Electric Pulp Testing (EPT)

Reliability & Clinical Notes

EPT checks the sensibility of a tooth by stimulating nerve endings with a low current and high potential difference in voltage. Although manufacturers give normal reference values of current, the best way to check “normal / baseline” values is to use the device on an adjacent, non-pathological tooth and compare the reading with the tooth in question.

EPT uses electrical excitation to stimulate the A-delta sensory fibres in the pulp. A positive response does not confirm the health or integrity of the pulp – it simply indicates that vital sensory fibres are present.

Important: EPT does not give any information about the vascular supply to the pulp, which is the true determinant of pulp vitality.

EPT is not considered reliable in the following conditions:

- A pus-filled canal — false positive
- A nervous patient — false positive
- Recent dental trauma — false negative
- Insulating restoration — false negative
- Secondary dentin deposits — false negative
- Moisture contamination — false positive
- Immature tooth (open apex) — false negative
- Patient who has taken analgesics — false negative
- Improper application / weak batteries in the EPT unit — false negative

Important: if a patient's medical history reveals a cardiac pacemaker, the use of an electric pulp tester is traditionally contraindicated.

Remember

- **Normal pulp:** a clinical diagnostic category in which the pulp is symptom-free and normally responsive to pulp testing.
- **Pulp necrosis:** a clinical diagnostic category indicating death of the dental pulp; the pulp is usually non-responsive to pulp testing.

Note: although previous international guidelines listed EPT as contraindicated in patients with cardiac pacemakers, it is not contraindicated with modern digital cardiac pacemakers, as these produce far less electrical interference.

Pulp Response to Microbial Invasion

MTA & Biodentine

Vital pulp (resistant / reversible pulpitis) is “fertile ground” for repair – vital pulp therapy (indirect or direct pulp capping) can be attempted. As invasion progresses to irreversible pulpitis and pulp necrosis, the tooth becomes non-vital and root canal treatment is required.

Remember

Pulp capping can be done in mechanical exposures; however, permanent teeth with a carious exposure generally require root canal treatment.

Mechanism of Action – MTA / Biodentine

- 1 MTA and Biodentine are highly biocompatible materials with an alkaline pH ($\approx 10.5\text{--}12$).
- 2 They promote hard-tissue repair by stimulating odontoblast activity and forming reparative dentin.

In past guidelines, calcium hydroxide was the material of choice for pulp capping. Current guidelines now recommend MTA and Biodentine as the first-choice pulp-capping materials.

First Drugs of Choice for Dental Diseases

According to ADA & ADC Guidelines

Includes maximum over-the-counter (OTC) dose and prescription dose for adults, unless otherwise stated.

Dental Condition	Drug of Choice	Mechanism / Purpose	Max OTC Dose (Adult)	Prescription Dose (Adult)
Dental pain (mild–moderate)	Ibuprofen (NSAID)	Analgesic, anti-inflammatory	400 mg every 6–8 h (max 1,200 mg/day)	400–800 mg every 6–8 h (max 2,400 mg/day)
	Paracetamol (acetaminophen)	Analgesic, antipyretic	1,000 mg every 6 h (max 4,000 mg/day)	1,000 mg every 6 h (max 4,000 mg/day)
Dental pain (combination)	Ibuprofen 400 mg + Paracetamol 1,000 mg	Synergistic analgesia	Every 6–8 h (max Ibuprofen 1,200 mg/day + Paracetamol 4,000 mg/day)	Ibuprofen 400–600 mg + Paracetamol 1,000 mg every 6–8 h (max Ibuprofen 2,400 mg/day + Paracetamol 4,000 mg/day)
Dental pain, severe (NSAIDs contraindicated)	Codeine + Paracetamol	Opioid analgesic (short-term use only)	NOT available OTC	Codeine 15–30 mg + Paracetamol 500–1,000 mg every 6 h (max Codeine 240 mg/day)
Dental infections	Amoxicillin	Antibiotic – first line for odontogenic infection	NOT available OTC	500 mg every 8 h (max 2,000 mg/day, usual 3–7 days)
Dental infection (anaerobic)	Metronidazole	Antibiotic for anaerobic infection	NOT available OTC	400 mg every 8 h (max 1,200 mg/day)
Gingivitis / plaque	Chlorhexidine 0.12% mouthwash	Antiseptic – reduces plaque & gingival inflammation	10–15 mL rinse twice daily (do not swallow)	Use >2 weeks only on advice; max duration without dental advice: 2 weeks
Oral thrush (candidiasis)	Nystatin oral suspension 100,000 units/mL	Antifungal	NOT available OTC	1 mL swish & swallow, 4x daily (usual 7–14 days)
Herpes simplex (labial)	Aciclovir	Antiviral – shortens course & severity	NOT available OTC	400 mg five times daily (usual 5 days)
Dry socket	Ibuprofen	Analgesic, anti-inflammatory	400 mg every 6–8 h (max 1,200 mg/day)	400–600 mg every 6–8 h (max 2,400 mg/day)
Aphthous ulcers (recurrent)	Triamcinolone 0.1% orabase	Topical corticosteroid – reduces pain & inflammation	NOT available OTC	Apply thin layer 2–4x daily after meals (avoid eating/drinking for 30 min)
Dental anxiety (pre-procedural)	Diazepam	Anxiolytic	NOT available OTC	5–10 mg, 1 hour before procedure (max 20 mg/day)
TMJ / muscle dysfunction	Ibuprofen or Naproxen	NSAID – pain & inflammation	Ibuprofen 400 mg every 6–8 h (max 1,200 mg/day) OR Naproxen 220 mg every 8–12 h (max 660 mg/day)	Ibuprofen 400–600 mg every 6–8 h (max 2,400 mg/day) OR Naproxen 250–500 mg every 12 h (max 1,000 mg/day)

General Principles (ADA & ADC)

- Use antibiotics only when there is evidence of infection or spread — not for pain alone.
- Analgesics are first-line for pain management; dose according to severity.
- Always consider medical history, allergies and drug interactions before prescribing.

Clinical Reminders

- Penicillin V (not amoxicillin) is generally regarded as the drug of choice for dental infection.
- Codeine is avoided for routine dental pain because of its sedative and addictive potential.
- A general dentist should not prescribe Nystatin for oral candidiasis on their own initiative – only an oral medicine specialist should, as it is broad-spectrum.

Note: most drugs listed here are not necessarily the definitive drug of choice for every individual scenario – included for knowledge purposes only. No patient-specific advice is given here.

Dens Invaginatus (Dens in Dente)

Developmental Anomaly

Dens invaginatus is a developmental anomaly caused by invagination of the enamel organ into the dental papilla before calcification.

Most Commonly Affected Tooth

- **Maxillary lateral incisor** (most common)

Less commonly affected:

- Maxillary central incisors
- Canines
- Premolars (rare)

Radiographic Appearance

“Tooth within a tooth” appearance.

Clinical Significance

- Deep, enamel-lined invagination
- Easy bacterial entry
- Early pulp necrosis
- A periapical lesion may develop even without obvious caries

Restorative Materials

Advantages & Disadvantages

Composite Resin

Advantages:

- Excellent aesthetics, shades and translucency
- Incremental placement builds anatomical form
- Modern fillers offer diverse physical properties
- High wear resistance for moderate load
- Strong micromechanical bond to enamel
- Dentine bonding agents help manage moisture and reduce sensitivity

Disadvantages:

- Placement is extremely technique-sensitive and requires high skill
- Contraction on setting (polymerisation shrinkage) creates stress at the tooth interface
- Risk of microleakage, which can lead to pulpal inflammation
- Requires carefully controlled moisture balance (avoid desiccation)
- Potentially shorter lifespan than some alternatives

Composite shrinkage: careful incremental placement helps limit the stress that can otherwise lead to pulpal inflammation and, in severe cases, pulp death.

Amalgam

Advantages:

- Relatively easy to handle
- Relatively tolerant of imperfect placement technique
- Excellent longevity in medium to extensive-sized lesions
- Relatively inexpensive

Disadvantages:

- Very poor aesthetics
- Requires undue sacrifice of sound tooth structure (retentive preparation)
- Difficult to restore full anatomical form
- Wear factor is too great for very extensive restorations

Glass Ionomer Cement

Advantages:

- Chemical (ion-exchange) adhesion to tooth structure
- Continuing fluoride reservoir
- Acceptable aesthetics
- Good wear resistance on maturity
- Low solubility on maturity

Disadvantages:

- Low fracture resistance
- Subject to dehydration in the absence of saliva

(Cast gold restorations are also referenced in the source material as a further comparison option, noted for the wide range of alloys available; further detail was not captured in the original note.)

Local Anesthesia in Dentistry

High-Yield Points

- 1 Fibres anaesthetised first during local anaesthesia: **C fibres**.
- 2 Local anaesthetic crosses the placenta through: **simple diffusion**.
- 3 Local anaesthetic used during epidural analgesia with the lowest fetal-to-maternal ratio: **Bupivacaine**.
- 4 Cardiovascular collapse occurs with: **Bupivacaine**.
- 5 Local anaesthetic with the highest protein-binding ability: **Bupivacaine**.
- 6 Short-acting local anaesthetic: **Procaine**.
- 7 Halothane leads to: **Bradycardia**.
- 8 Local anaesthetic with the highest oxygen-binding: **Prilocaine**.
- 9 Local anaesthetic acting for more than 2 hours: **Bupivacaine**.
- 10 Topical use of this local anaesthetic is not required: **Bupivacaine**.
- 11 Most cardiotoxic local anaesthetic: **Bupivacaine**.
- 12 Cardiac resuscitation is toughest after use of: **Bupivacaine**.

(The source note continues with "anaesthetic of choice used in epidural..." but the message was cut off in the original screenshot.)

Imaging of the Oro-Facial Complex

High-Yield Points

- 1 **Intra-oral radiographs** – Anatomical structures: dentoalveolar complex. Diseases: dental caries and its sequelae, periodontal disease.
- 2 **Extra-oral radiographs** – Anatomical structures: skull, sinonasal complex, jaws. Diseases: odontogenic cysts/neoplasms, bone diseases, sinusitis.
- 3 **Cone Beam Computed Tomography (CBCT)** – Anatomical structures: dentoalveolar complex, maxillary antrum. Diseases: odontogenic cysts/neoplasms, bone diseases, sinusitis.
- 4 **Ultrasound** – Anatomical structures: major salivary glands, cervical lymph nodes, thyroid gland, soft tissues of the neck. Diseases: sialolithiasis/sialadenitis, lymphadenopathy, thyroiditis, thyroid neoplasia.
- 5 **Sialography** – Anatomical structures: submandibular gland, parotid gland. Diseases: sialolithiasis/sialadenitis.
- 6 **Computerized Tomography (CT)** – Anatomical structures: head and neck tissues. Diseases: mainly used for neoplasia.
- 7 **Magnetic Resonance Imaging (MRI)** – Anatomical structures: head and neck tissues. Diseases: mainly used for neoplasia.
- 8 **Positron Emission Tomography–Computed Tomography (PET-CT)** – used mainly in oncological imaging.

(Item 8 was cut off at the end of the original message; the source did not include further detail beyond the PET-CT entry.)



The Dentistry Clinical Picture Gallery

Simple picture-based representations of clinical conditions

Reference: Australian Dental Therapeutic Guidelines

NON CARIOUS LESIONS



ATTRITION

- **Tooth-to-tooth wear.**
- **Happens in bruxism, aging, parafunction.**
- **Flat facets, matching occlusal surfaces.**
- **May expose dentin, sensitivity present.**

- **Chemical dissolution without bacteria.**
- **Acidic diet, GERD, vomiting.**
- **Smooth, saucer-shaped, shiny enamel loss.**
- **Affects palatal maxillary incisors (intrinsic) / buccal (extrinsic).**



EROSION



ABFRACTION

- **Stress-induced cervical microfractures.**
- **Due to occlusal overload + flexure.**
- **Wedge-shaped, sharp margins.**
- **Often isolated to one tooth under heavy load.**

- **Mechanical wear by external objects.**
- **Common causes: hard brushing, toothpicks, habits.**
- **V-shaped notches at cervical area.**
- **Canines, premolars affected mostly.**



ABRASION

Clinical Reference Image

Australian Dental Therapeutic Guidelines

EPINEPHRINE DRUG CLASS: ALPHA/BETA ADRENERGIC AGONISTS (CONSIDERED A VASOPRESSOR) USE: - VF/Pulseless VT - Symptomatic bradycardia - Anaphylaxis - Asystole/PEA PATHOPHYSIOLOGY: Stimulates alpha and beta-adrenergic receptors to relax heart & lung smooth muscle & dilate vessels DOSING: VF/pVT/PEA/Asystole: 1mg IV/IO q3-5min Think: "push 1 of Epi" Symptomatic Bradycardia: Start infusion at 2-10 mcg/min IV/IO titrating to the patient's response 	ATROPINE DRUG CLASS: ANTICHOLINERGIC USE: - Bradycardia - Other: Surgery (to reduce saliva, mucus, or other secretions in the airway) PATHOPHYSIOLOGY: "Anti-" = "stops" + Cholinergic = acetylcholine AKA stops acetylcholine from binding to receptors, which increases heart rate! DOSING: 1 mg IV/IO bolus q3-5min with a max dose of 3mg 
AMIODARONE DRUG CLASS: ANTIARRHYTHMIC USE: - VF/Pulseless VT - Recurrent, hemodynamically unstable VT PATHOPHYSIOLOGY: Blocks potassium currents that normally cause repolarization of heart muscle. This will reduce cardiac muscle excitability, lower the heart rate and "hopefully" slow down the heart! DOSING: 300 mg IV/IO push; second dose 150 mg IV/IO push AMI SLOW DARONE  SLOWING DOWN THE HEART RATE!	ADENOSINE DRUG CLASS: ANTIARRHYTHMIC USE: When vagal maneuvers fail to terminate stable narrow-complex SVT or VT PATHOPHYSIOLOGY: Momentarily slows down conduction in the AV node by interrupting the AV nodal circuit and converting SVT into "hopefully" sinus rhythm! DOSING: FIRST DOSE: 6mg IV/IO RAPID push over 1-3 sec followed by RAPID 20ml flush & elevation of extremities SECOND DOSE: 12mg IV/IO RAPID push over 1-3 seconds 
CALCIUM CHLORIDE DRUG CLASS: ELECTROLYTE USE: - Cardiac arrest - Hyperkalemia - Hypocalcemia PATHOPHYSIOLOGY: "C"ontraction of major muscles/organs like the heart DOSING: IN CARDIAC ARREST: 20 mg/kg IV/IO bolus IN NON-ARREST: Infuse over 30-60 minutes 	SODIUM BICARB DRUG CLASS: ALKALINIZING AGENT USE: - Tricyclic antidepressant overdose - Corrects metabolic acidosis in cardiac arrest (Acid wastes build up in the body after the heart stops beating!) - Cardiac arrest due to hyperkalemia DOSING: 1 mEq/kg IV push in a dose of 50ml D5W  @stephaneebeggs
MAGNESIUM SULFATE DRUG CLASS: ANTIARRHYTHMIC AND ELECTROLYTE USE: - Terminate or prevent recurrent VT associated with torsades de pointes - Arrhythmias related to hypomagnesemia PATHOPHYSIOLOGY: "M"uscle relaxation in major organs = slows down the impulse rate of the SA node to relax heart rate DOSING: 1-2g diluted in 50-100ml D5W IV push over 5-20 minutes 	LIDOCAINE DRUG CLASS: ANTIARRHYTHMIC USE: - In arrests due to: - Vfib - Pulseless ventricular tachycardia - Local anesthesia PATHOPHYSIOLOGY: Slows the influx of sodium, raising the cells' electrical stimulation threshold and restoring cardiac rhythm 

HYPOGLYCAEMIA MANAGEMENT IN DENTAL CHAIR

Recognize Early • Act Quickly • Prevent Complications

1. RECOGNIZE HYPOGLYCAEMIA

- ✓ Sweating, cold clammy skin
- ✓ Shakiness, tremor
- ✓ Paleness
- ✓ Hunger
- ✓ Anxiety, irritability, confusion
- ✓ Dizziness, blurred vision
- ✓ Headache
- ✓ Tachycardia
- ✓ In severe cases: weakness, drowsiness, seizures, loss of consciousness



RISK FACTORS

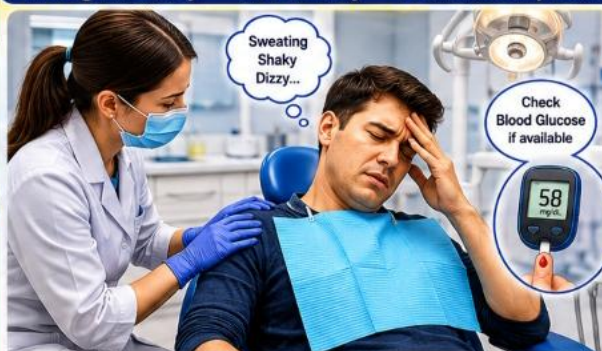
- Skipped meal
- Excess insulin or oral hypoglycaemic drugs
- Extra physical activity
- Long dental appointments
- Alcohol intake

2. INITIAL ACTIONS

- STOP** Stop dental treatment immediately
- Sit the patient upright (if possible)
- Assess ABC and level of consciousness
- Check Blood Glucose if available
- Reassure the patient and keep calm

WHY NOT GIVE NITROGLYCERIN TO UNPRESCRIBED PATIENTS?

- ✗ Chest pain may not always represent stable angina
- ✗ Undiagnosed ischemic heart disease
- ✗ Hypotension and drug interactions (e.g., sildenafil/PDE-5 inhibitors)
- ✗ Evolving myocardial infarction can complicate GTN use
- ✗ Dental clinics are not intended for initiating antianginal therapy



3B. MANAGEMENT – PATIENT UNCONSCIOUS (Or unable to swallow)

- ✓ Do NOT give anything orally
- ✓ Place in recovery position
- ✓ Airway, Breathing, Circulation – check
- ✓ Give high flow oxygen (6–10 L/min) if available
- ✓ If trained and available: Give Glucagon 1 mg IM/SC
- ✓ Monitor vital signs
- ✓ Call Emergency Medical Services (000)
- ✓ Transfer to hospital



3A. MANAGEMENT – PATIENT CONSCIOUS

- ✓ Stop dental procedure immediately
- ✓ Sit the patient upright
- ✓ Reassure and Talk calmly
- ✓ Give FAST ACTING CARBOHYDRATE (15 g):



- ✓ Recheck after 15 minutes
- ✓ If symptoms persist, repeat carbohydrate
- ✓ When improved, give a snack (complex carbohydrate)
- ✓ Postpone further treatment
- ✓ Document the event

Use of NITROGLYCERIN (GTN) in Hypoglycaemia?
Not indicated. Chest discomfort in hypoglycaemia is usually due to autonomic symptoms, not angina.

★ IMPORTANT POINT

- Different international dental guidelines have different recommendations regarding nitroglycerin administration in angina patients who were not previously prescribed nitroglycerin.
- According to Australian Therapeutic Guidelines (Oral & Dental), nitroglycerin should **NOT** be administered in the dental clinic to a patient who has not previously been prescribed it.
- If the patient has not been prescribed nitroglycerin, perform all other emergency management steps except nitroglycerin administration.

4. MONITOR & FOLLOW UP

- Monitor vital signs
- Observe until fully recovered
- Document the event
- Advise patient to eat regular meals & carry glucose source

5. WHEN TO CALL EMERGENCY SERVICES

- If unconscious or seizing
- If no improvement after treatment
- If repeated episodes or underlying medical concern
- Any doubt about the patient's condition



Remember: Early recognition + Prompt treatment = Prevention of serious complications

Clinical Reference Image

Australian Dental Therapeutic Guidelines

EMERGENCY MANAGEMENT OF PULMONARY EMBOLISM IN DENTAL CHAIR

1. RECOGNISE



- Sudden shortness of breath
- Sharp pleuritic chest pain
- Tachycardia (>100/min)
- Tachypnea
- Anxiety, restlessness
- Drop in SpO₂ (<94%)
- Syncope / collapse ±



2. IMMEDIATE ACTIONS



Stop dental procedure immediately



Call for help / Medical emergency team



Sit patient upright / semi-reclined



Ensure open airway



Loosen tight clothing



Reassure and stay calm

SUDDEN

- Dyspnea
- Pleuritic chest pain
- Tachycardia
- Tachypnea
- Anxiety
- Syncope ±

3. OXYGEN THERAPY



- Give high flow oxygen 10–15 L/min via non-rebreather mask



- Target SpO₂ ≥ 94%
- Monitor SpO₂ continuously

4. MEDICATION



If skilled & authorised:

Adrenaline 0.5 mg IM (1:1000) if anaphylaxis is suspected



Aspirin 300 mg chewed (if no contraindication)



Consider GTN 0.4 mg SL (only if SBP > 100 mmHg and no RV infarction)



IV access – Normal saline

5. MONITOR & SUPPORT



- Monitor Pulse, BP, RR, SpO₂, ECG



- Look for shock signs



- Keep patient warm

- Do not leave patient alone

6. WHEN TO CALL FOR HELP



Call emergency medical service immediately if:

- SpO₂ < 94% despite oxygen
- Severe breathlessness
- Chest pain persists
- Hypotension / Syncope
- Suspected PE / DVT

7. PREPARE FOR TRANSFER



- Provide continuous oxygen
- Keep patient in semi-reclined position
- Monitor during transfer
- Send with medical notes & medication history

IMPORTANT POINTS



Do not give sedatives



Avoid vasoconstrictor (Adrenaline in LA)



Avoid unnecessary movement



Treat as medical emergency

RISK FACTORS (REMEMBER)



Recent surgery



Immobilization



Oral contraceptives



Smoking



Obesity

Clinical Reference Image

Australian Dental Therapeutic Guidelines

1. ASSESS & RECOGNISE

- Sudden shortness of breath
- Pleuritic chest pain
- Rapid breathing
- Anxiety, sense of impending doom
- Tachycardia
- Hypotension (late sign)
- Low oxygen saturation
- Risk factors: recent surgery, immobilization, OCP use, pregnancy, DVT, cancer, thrombophilia



EMERGENCY MANAGEMENT OF PULMONARY EMBOLISM IN DENTAL CHAIR

ASSESS – RECOGNISE – RESPOND



2. IMMEDIATE ACTIONS

- Stop dental treatment immediately
- Reassure the patient and stay calm
- Position semi-reclined or upright (as tolerated)
- Ensure open airway
- Loosen tight clothing

3. OXYGEN THERAPY

- Administer oxygen immediately
- High flow: 10–15 L/min via non-rebreather mask
- Aim to maintain $SpO_2 \geq 94\%$

4. MONITOR & SUPPORT

- Monitor vital signs:
 - Pulse, BP, RR, SpO_2
- Continuous observation
- Provide reassurance and emotional support
- Be prepared for deterioration

5. MEDICATION (IF TRAINED & AUTHORISED)

- Anticoagulant
(Do not give unless trained and authorised)
- If hemodynamically unstable and arrest imminent:
 - Adrenaline 0.5 mg IM (1:1000)
 - Repeat every 3–5 min if needed
- If severe hypotension and trained:
 - IV fluid bolus (0.9% Normal Saline)

6. WHEN TO CALL FOR HELP

- Call emergency medical services immediately if:
 - Sudden breathlessness
 - $SpO_2 < 94\%$ despite oxygen
 - Chest pain + tachycardia
 - Collapse or hypotension
 - Suspected pulmonary embolism

7. PREPARE FOR TRANSFER

- Arrange urgent transfer to hospital
- Keep patient on oxygen
- Provide details of events and management
- Send with medical history and medications

ADDITIONAL CONSIDERATIONS

- Medical history review:
 - Previous DVT/PE
 - Clotting disorders
 - Medications (e.g., anticoagulants)
- Postpone elective dental treatment until recovery and medical clearance
- Use standard infection control precautions
- Document the event, management and patient response

EARLY RECOGNITION AND PROMPT MANAGEMENT CAN SAVE LIVES

MANAGEMENT OF THE DIABETIC PATIENT IN THE DENTAL CHAIR

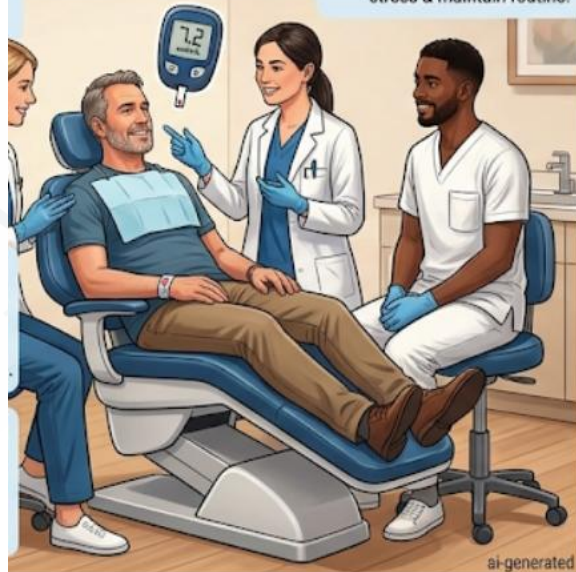
ASSESSMENT & PREVENTION

2 ASK ABOUT RECENT BLOOD GLUCOSE

Has the patient eaten and taken medication as usual?

3 CONSIDER TREATMENT TIMING

Prefer short morning appointments to reduce stress & maintain routine.



INTRAOPERATIVE & EMERGENCY MANAGEMENT

1 IF SYMPTOMS OF HYPOGLYCEMIA OCCUR

(e.g., shaking, confusion, sweating)



Stop treatment immediately.

3 MONITOR & REPEAT

Check with the patient's glucometer if handy.

5 IF SEVERE / LOSS OF CONSCIOUSNESS



Administer IV glucose or glucagon if trained, or call 000 immediately.

2 CONFIRM BG IF POSSIBLE

Check with the patient's glucometer if handy.

3 ADMINISTER 15g FAST-ACTING CARBOHYDRATE



*Only if conscious

4 MONITOR & REPEAT

Wait 15 mins.
Re-check BG.
Repeat if still low.

6 RECORD VITAL SIGNS



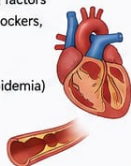
Heart monitor.
Record event details.

MANAGEMENT OF ANGINA PECTORIS PATIENT IN DENTISTRY

Goal: Provide safe dental care, prevent angina attack, and manage it promptly if it occurs.

1. PRE-TREATMENT ASSESSMENT

- Detailed medical history
 - Type of angina, frequency, precipitating factors
 - Current medications (e.g., nitrates, β -blockers, CCBs, antiplatelets)
 - Last episode, exercise tolerance
 - Associated conditions (HTN, DM, dyslipidemia)
- Consultation with physician if needed
- Vital signs: BP, Pulse, SpO₂
- Classify severity (CCS I-IV)



2. PREVENTION – REDUCE RISK OF ANGINA

- Stress reduction: calm environment, short appointments
- Adequate local anesthesia – infiltrative > block
- Use vasoconstrictor with caution
 - Prefer 1:100,000 epinephrine (limit dose)
 - Avoid 1:80,000 in severe/uncontrolled angina
- Avoid hypoxia: good ventilation, oxygen if needed
- Avoid extremes of BP and HR
- Profound anesthesia to minimize pain & anxiety
- Do not discontinue cardiac medications
- Morning appointments preferred
- Limit physical exertion



3. DRUG CONSIDERATIONS

Safe / Preferred	Use with Caution / Avoid
✓ Acetaminophen	✗ NSAIDs (may increase CV risk)
✓ Local anesthetics with 1:100,000 epinephrine (≤ 2 cartridges)	✗ Tricyclic antidepressants
✓ Amoxicillin, Clindamycin, Azithromycin (if antibiotic needed)	✗ High doses of epinephrine (vasoconstrictor)
	✗ Excessive sedation



4. IF ANGINA OCCURS IN DENTAL OFFICE

- Stop treatment immediately
- Sit the patient upright, reassure and calm
- Administer oxygen 4–6 L/min
- Give patient's own GTN (glyceryl trinitrate)
 - 0.4 mg sublingual spray/tablet
 - Repeat every 5 minutes (max 3 doses)
- Monitor: BP, Pulse, SpO₂
- If pain persists > 15 minutes or worsens, or associated with breathlessness, sweating, nausea → Activate emergency medical services



5. EMERGENCY MANAGEMENT (IF NO RELIEF)

- Continue oxygen
- Aspirin 150–325 mg to chew (if not allergic and not contraindicated)
- GTN as above
- Monitor vitals
- Urgent referral to hospital (suspected myocardial infarction until proven otherwise)



6. POST-TREATMENT

- Ensure complete recovery before discharge
- Advise rest and avoid strenuous activity
- Document the event and management
- Communicate with patient's physician



7. KEY POINTS TO REMEMBER

- Prevention is the best approach
- Maintain adequate oxygenation, pain control, and hemodynamic stability
- Always have emergency drugs and equipment available
- Early recognition and prompt management can save life

Common Patient Medications



GTN (Sublingual)



β -Blockers



CCBs



Antiplatelets
(Aspirin/Clopidogrel)

Emergency Kit in Dental Office

- ✓ Oxygen
- ✓ GTN spray/tablets
- ✓ Aspirin
- ✓ BP monitor
- ✓ Stethoscope
- ✓ Emergency contact numbers

Types of Dental Bridges



Traditional Bridge



Cantilever Bridge



Maryland Bridge



Implant-Supported Bridge

Drug Antidotes

Antidote	Indication
◦ Physostigmine	◦ Anticholinergic poisoning
◦ Protamine sulfate	◦ Heparin poisoning
◦ N-acetylcysteine	◦ Acetaminophen toxicity
◦ Pyridoxine	◦ Isoniazid toxicity
◦ Naloxone	◦ Opioid overdose
◦ Leucovorin	◦ Methotrexate toxicity
◦ Flumazenil	◦ Benzodiazepine toxicity
◦ Deferoxamine	◦ Iron poisoning
◦ EDTA; dimercaprol	◦ Heavy metal poisoning
◦ Succimer / DMSA	◦ Lead poisoning
◦ Activated charcoal	◦ Used for many oral toxins
◦ Idarucizumab	◦ Reversal of dabigatran
◦ Phytomenadione	◦ Warfarin overdose
◦ Sodium bicarbonate	◦ Aspirin toxicity
◦ Andexanet alfa	◦ Apixaban / rivaroxaban toxicity
◦ Glucagon	◦ Beta-blocker toxicity
◦ Ethanol	◦ Ethylene glycol / methanol poisoning
◦ Prussian blue	◦ Thallium / cesium poisoning

Comparison between RUBBER BASE impression materials

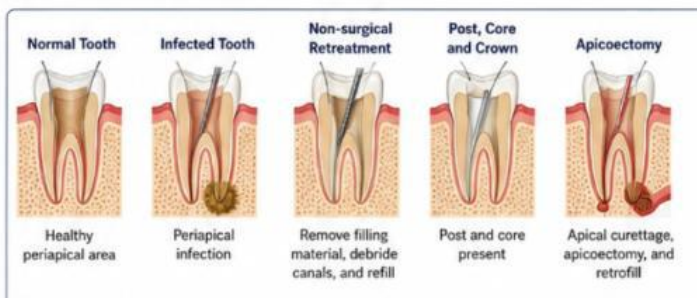
AZHAR ASSUIT		Cond. Silicon	Addition Silicon	Poly ether	Poly sulfide
Composition		80% Silicon (OH)group 20% Silica	Tin octate Ethel silicate 80%Silicon (vinyl)group 20% Silica	80% polyether (ethylene imine) 20% Silica	80% polysulfide (SH)group 20% Silica
Fluidity		Depend on consistency القوام by filler as silica:- types & route of application العينة المستخدم لتطبيع العجينة 1- Light body → Syringe 2- Medium → Tray 3- Heavy → Tray + Syringe			
Accuracy		Increase with 2nd imp.	Highest accuracy	More accuracy	Accurate
Tray & spacer thickness		Should be acrylic tray -- Impression material thickness 2-4 mm			
Dimensional change		0.6%	.05 %	0.1 %	.45% alcoholic evaporation
Shrinkage degree		0.2-1% in 24hrs	Lowest shrinkage	.3% in 24hrs	.3-.4 %
Elastic recovery		99%	99.5%	98.5 %	96 %
Properties	1. Handle mixing	1. Easy mixing	1. Very easy mixing	1. Moderate mixing	1. Moderate - difficult
	2. Pouring ease	2. Poor to adequate	2. Adequate in stone	2. Easily poured in stone	2. Adequate pour
	3. Pouring time	3. Once أو 1 hour	3. 60min- 1 week	3. (1) week	3. 1 hour
	4. Tear strength	4. Adequate tear St.	4. Adequate tear St.	4. Adequate tear St.	4. Highest tear St.
	5. Cost / taste	5. Suitable حزين	5. High cost	5. High cost -Bitter taste	5. Bad odder
	6. Working time	6. Long 10 min	6. Short 6-8 min	6. Short 5-6 min	6. Long 12-14 min
	7. Hydrophilic or	7. Very hydrophobic	7. Hydrophilic	7. Distort if stored in water (hydrophilic)	7. Hydrophilic
	8. others	8. Less distortion	8. Doesn't stain NB: should poured after setting at least 30-60 min due to elastic recovery	8. Allergic reaction (sulfonic acid)	8. Clothing stain - Condensation reaction

APICOECTOMY

An apicoectomy is the preparation of a flat surface by the excision of the apical portion of the root and any subsequent removal of attached soft tissues.

If a tooth has had previous endodontic therapy and becomes reinfected, it is usually best to try and retreat it conventionally — remove filling material, debride the canals, and refill. However, if the tooth has been restored with a post, core, and crown, then apical curettage, apicoectomy, and a retrofill should be performed.

Note: Retreating a tooth with a post is the most common reason for an apicoectomy and retrograde filling.



INDICATIONS FOR APICOECTOMY (ROOT-END RESECTION):

- A reverse filling needs to be placed
- It is necessary to gain access to an area of pathosis
- The poorly filled apical portion of the root is to be removed to the level of canal obliteration
- Non-negotiable canal, blockage, or severe root curvature in which nonsurgical treatment is impossible
- Complications arising from procedural accidents (e.g., separation of instruments, ledging, and/or perforations) which cannot be handled without surgical exposure of the site
- Failed treatment due to irretrievable posts or root fillings
- Horizontal apical fractures in which apical end of the pulp becomes necrotic
- Biopsy — to diagnose non-odontogenic causes of symptoms (e.g., patient with a history of previous malignancy, lip paresthesia, or anesthesia)

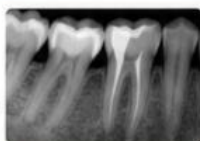
CONTRAINDICATIONS FOR APICOECTOMY (ROOT-END RESECTION):

- Anatomic factors that limit access
- Medical or systemic complications
- Tooth is nonrestorable or has a poor crown/root ratio

PROCEDURE:

1 RADIOGRAPHS

Radiographs are taken to determine the length of the root and its proximity to adjacent structures.



2 ADMINISTER ANESTHESIA

Local anesthesia is administered.



3 ON THE LABIAL SURFACE OF THE TOOTH,

with the help of a periosteal elevator, locate the root apex, so that an incision can be made.



4 FLAP DESIGNS USED:

- Submarginal scalloped (Ochsenbein-Luebke)
- Full mucoperiosteal flaps (the best)



5 REFLECT THE FLAP



6 ROOT APEX IS EXPOSED,

then apex is cut off with a fissure bur about one-third of its length.



7 CURETTE

the surrounding pathologic tissues and round off the end of the cut root.



8 FOR RETROGRADE FILLING,

a bevel of 0–10 degrees is given.



9 RETROGRADE FILLING

to 3 mm is done.



10 IRRIGATE THE WOUND AND SUTURE THE FLAP IN POSITION



SUMMARY OF APICOECTOMY STEPS



REMEMBER: Retreating a tooth with a post is the most common reason for an apicoectomy and retrograde filling.



RADIATION SAFETY AND DIAGNOSTIC RADIOGRAPHS IN DENTISTRY



The goal of radiation protection is to get the best diagnostic image with the least radiation exposure.

NOTES RELATED TO RADIATION SAFETY AND DIAGNOSTIC RADIOGRAPHS

1. A fast (sensitive) film, for example F-speed film, is preferred over both E- and D-speed films. Faster films can significantly cut down on the amount of radiation required for a quality image.
2. Dental units should operate at **70 kV or higher**. The higher the kV, the lower the patient's skin doses. **Note:** The optimal setting for maximal contrast between radiopaque and radiolucent structures is **70 kV**.
3. **Collimation** (i.e., restriction of the x-ray beam size so that it does not exceed 2.5 inches at the patient's skin, reduces exposure).
4. Patient should be protected with a lead apron and a thyroid collar for each exposure.
5. If there is no barrier for the clinician to stand behind while exposing films, he/she should stand in an area of **minimal scatter radiation** (i.e., 6 feet away and in the area that lies between 90° to 135° to x-ray beam).
6. **Dental personnel** who may get exposed to occupational x-radiation **must wear film badges** to record exposure and **must never exceed the maximum permissible dose (MPD) of 0.05 Sv/year**.
7. An operator should **never remain** in the room holding an x-ray packet in place for a patient. If film must be held in place by someone (i.e., for a child), drape the patient and have him/her hold the film.
8. The **most accurate** radiographs for endodontics are made using the **paralleling technique**. **Remember:** When using the paralleling technique, you must center the x-ray film packet behind and parallel with the long axis of the tooth being x-rayed. The tube head must be positioned so that the central x-ray beam is projected perpendicular to the tooth and the film packet.

RADIATION PROTECTION PRINCIPLES

TIME	DISTANCE	SHIELDING	COLLIMATION
Use the shortest exposure time possible.	Maximize distance between source and operator.	Use barriers and protective devices (lead apron, thyroid collar).	Restrict the beam size to the area of interest.

PERSONAL PROTECTIVE EQUIPMENT

Lead apron (0.25–0.35 mm Pb)	Thyroid collar (0.5 mm Pb)	Protective eyewear (lead glasses)

OCCUPATIONAL EXPOSURE MONITORING

Wear film badge dosimeter at collar level (outside the apron).



Maximum Permissible Dose (MPD) for occupational exposure:

0.05 Sv/year
(5 rem/year)

SAFE OPERATING GUIDELINES

Never remain in the room holding the film packet. 	Use proper patient positioning and stabilization (holders, bite blocks). 	Drape the patient with a lead apron. 	Use rectangular collimation (preferably). 	Maintain the recommended distance (6 feet or more) from the source. 	Keep the equipment well maintained and exposed films/sensors properly processed.
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PARALLELING TECHNIQUE (LONG CONE)



Principle: The x-ray beam is directed perpendicular (90°) to both the long axis of the tooth and the image receptor (film/PSP/sensor).

STEPS

1. Position the receptor (film/PSP) parallel to the long axis of the tooth.
2. Direct the central ray perpendicular (90°) to the receptor.
3. Center the beam through the contact area.

ADVANTAGES

- Most accurate representation
- Minimal distortion and magnification
- Ideal for all intraoral radiographs

CLINICAL USES

- Periapical radiographs (anterior & posterior)
- Bitewing radiographs
- Endodontic, periodontal, and implant evaluation

ADDITIONAL GOOD PRACTICES

Use fastest film/sensor available. 	Use highest kV (≥ 70 kV) for lower dose. 70kV or higher	Use proper exposure time (lowest possible time for your system). 	Use digital sensors when available – lower dose and instant results. 	Keep children's exposures as low as reasonably achievable. 	Pregnant women: expose only if essential.
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MAXIMIZE DIAGNOSTIC INFORMATION

- ✓ Correct patient positioning
- ✓ Proper angulation
- ✓ Proper exposure
- ✓ Proper processing / image viewing
- ✓ Repeat only if necessary

COMMON INTRAORAL X-RAY TECHNIQUES (OVERVIEW)

BISECTING ANGLE (SHORT CONE) Vertical angulation down the bisector. • Used mainly for children • More distortion	OCCCLUSAL TECHNIQUE Perpendicular to occlusal plane. • Occlusal radiographs • Palatal/impacted teeth	MAXILLARY SINUS (WATERS' VIEW) Through maxillary sinus. • Detect sinus pathology, cysts, tumors	MANDIBULAR CANAL (TOWNE'S VIEW) 20°–30° caudally. • Mandibular canal, fractures, impacted teeth	TMJ (LATERAL JAW) VIEW 2.5 cm anterior to external auditory meatus. • Evaluate condyle, fossa, joint space
--	--	--	--	---

POTENTIAL HAZARDS OF RADIATION



- Skin reactions, erythema
- Cataracts (high exposure)
- Genetic effects
- Increased risk of cancer (very high cumulative dose)

ALARA PRINCIPLE:
As Low As Reasonably Achievable



Good Radiographs = Accurate Diagnosis = Better Treatment Outcomes



Clinical Reference Image

Australian Dental Therapeutic Guidelines



DMFT Index

What it stands for

Letter	Meaning
D	Decayed
M	Missing (due to caries)
F	Filled
T	Teeth

Simple Definition



DMFT is a scoring system used to measure the caries experience of an individual or a population. It tells you how many permanent teeth have been decayed, are missing due to decay, or have been filled.

How it's calculated

- 1 Examine all 32 permanent teeth (or 28, excluding third molars)
- 2 Each tooth gets a score of D, M, or F
- 3 Add them all up → that's the DMFT score

Example: 2 decayed + 1 missing + 3 filled = DMFT of 6

For PRIMARY Teeth → dmft (lowercase)

Letter	Meaning
d	Decayed
m	Missing
f	Filled
t	Teeth



Written in lowercase to distinguish from permanent dentition

Key Rules to Remember

-  Missing teeth are only counted as M if lost due to caries — not trauma or ortho extraction
-  A tooth that is both decayed and filled → counted as **D only**
-  Unerupted teeth are excluded
-  Third molars are often excluded unless specified
-  Each tooth scored once only

DMFS — Surface Level Version



- Instead of teeth, counts surfaces:
- Posterior teeth → 5 surfaces each
- Anterior teeth → 4 surfaces each
- More sensitive than DMFT for detecting early caries

WHO Caries Severity Scale (DMFT scores)

DMFT Score	Severity
0	Caries free
1–3	Low
4–6	Moderate
7–9	High
≥10	Very high

Why it matters for ADC



Used in community dentistry / public health questions

Traps



Common trap: counting a traumatically lost tooth as M — wrong



Common trap: scoring a D+F tooth as F — wrong, always D

Topical Fluoride Gel or Foam



Professionally applied topical fluoride gel is an effective way of preventing dental decay in patients at high-risk of developing caries. However, **high concentration fluoride gels and foams (>1.5mg/g F ion) are contra-indicated for use in children under 10 years of age due to the potential for ingestion.** Furthermore, fluoride gels appear more efficacious on the permanent dentition.

¹ Under review, 2009

Page 1 of 7



dental health
services victoria
oral health for better health

There are two types in common usage: **1.23% Acidulated Phosphate Fluoride (APF)** and **2.0% Neutral Sodium Fluoride (NaF)** (9,000 parts per million (ppm) F⁻). Due to its neutral pH, NaF is now more widely used. It is particularly indicated for use in patients with the following:



- Composite resin, glass ionomer or porcelain restorations, as the acidity of the APF gels, if used 2–3 monthly, can cause damage to these restorations



- Enamel erosion



- Very porous enamel structure such as carious or hypoplastic enamel



- Exposed or carious dentine and root surfaces



Fluoride foams for topical application are also available however, there are few clinical studies of the effectiveness of fluoride foams.



Topical fluoride Gels and foams used to be applied in dental trays, but this practice is obsolete, with the advent of fluoride varnishes.

Spot Application



Spot application of fluoride gels may be appropriate for patients with limited co-operation.

APF is used for spot fluoride application (unless NaF is indicated) and applied for **four minutes**.

The patient is given the same instructions as they receive after a full mouth application.

Fluoride Varnish



Duraphat™, a 5% NaF varnish (22,600 ppm F⁻), is used for spot applications on enamel caries diagnosed visually or by radiograph, rather than for whole mouth application.

It must be used with care in minute quantities (deciduous dentition 0.25 ml, mixed dentition 0.40 ml, permanent dentition 0.75 ml) due to its high fluoride concentration. Duraphat's™ effect is similar to that obtained with other topical fluoride preparations but is less efficient due to its high cost. An indication for its use is when treating patients who have elevated risk of developing caries, e.g. limited co-operation due to disabilities, and including children less than ten years of age.



Patients with an allergy to Colophony should not use Fluoride Varnishes.



Clinical Reference Image

Australian Dental Therapeutic Guidelines



COMPLETE CROWN GUIDE

Indications • Preparation • Materials • Advantages • Disadvantages

1. FULL METAL CROWN (CAST METAL)

Materials

Noble alloys: gold (Type III/IV), gold-palladium

Base metal alloys: Ni-Cr, Co-Cr

Preparation Design

- Occlusal reduction: 1–1.5mm
- Axial reduction: 0.5–1mm
- Margin: Chamfer or knife-edge (feather edge)
- Taper: 6° total (3° each side)
- Surface area: Maximum retention – least tooth structure removed

✓ Indications

- Posterior teeth (molars/premolars)
- Limited interocclusal space (<1.5mm)
- Bruxism / heavy occlusal forces
- Short clinical crowns
- Abutment for FPD
- Subgingival caries extending below CEJ

✗ Contraindications

- Anterior/visible teeth
- Metal allergy (Ni-Cr)
- Adequate space for all-ceramic alternative
- Patient preference against metal

✓ Advantages

- Best marginal fit of all crowns
- Most durable – highest fracture resistance
- Least tooth reduction required
- Biocompatible (gold alloys)

✗ Disadvantages

- Poor aesthetics
- Ni-Cr: potential allergy/galvanism
- Gold: expensive

2. PORCELAIN-FUSED-TO-METAL (PFM)

Materials

Metal coping (Ni-Cr, Co-Cr, gold-palladium) + feldspathic porcelain baked over

Preparation Design

- Occlusal reduction: 1.5–2mm
- Axial reduction: 1–1.5mm
- Margin:
 - Porcelain margin (all-ceramic margin): shoulder or deep chamfer – 90° butt joint, 1–1.5mm
 - Metal collar margin: knife-edge or chamfer
- Shoulder for porcelain area must be flat to support porcelain
- Taper: 6° total

✓ Indications

- Anterior AND posterior teeth
- Long-span bridges
- Abutments for FPD
- Moderate-to-heavy occlusal load with aesthetic requirement
- When all-ceramic insufficient strength

✗ Contraindications

- Very short clinical crowns
- High smile line / recession risk
- Proven metal allergy
- Parafunctional – porcelain fracture risk
- Limited interocclusal space <1.5mm

✓ Advantages

- Good aesthetics + strength
- Long clinical track record
- Can be used for long-span bridges

✗ Disadvantages

- Porcelain fracture
- Metal show at margin (grey line)
- Gingival discolouration (Ni-Cr)
- More tooth reduction required
- Opaque appearance
- Requires porcelain shoulder prep

3. ALL-CERAMIC CROWNS

	3a. FELDSPATHIC PORCELAIN	3b. LEUCITE (IPS EMPRESS)	3c. LITHIUM DISILICATE (IPS e.max)	3d. ZIRCONIA (Y-TZP)
Material	Pure silica-based glass ceramic	Leucite crystals dispersed in glass matrix (heat-pressed)	Lithium disilicate crystals in glass (heat-pressed or CAD/CAM milled)	Polycrystalline ceramic (no glass phase – CAD/CAM only)
Preparation	Occlusal: 1.5–2mm Axial: 1–1.5mm Margin: Shoulder (90°), 1–1.5mm	Occlusal: 1.5–2mm Axial: 1–1.5mm Margin: Shoulder (90°), 1–1.5mm	Occlusal: 1–1.5mm (min 1mm CAD/CAM) Axial: 1mm Margin: Chamfer or shoulder, 1mm	Occlusal: 0.5–1.5mm (min 0.5mm mono.) Axial: 1mm Margin: Chamfer 0.5–1mm
Indications	• Anterior teeth • Maximum aesthetics • Veneers • Low occlusal load	• Anterior crowns • Veneers • Inlays/onlays • 3-unit FPD (anterior only, limited)	• Anterior & premolar crowns • Inlays, onlays, veneers • Implant crowns (ant./premolar) • 3-unit FPD up to second premolar	• Posterior crowns • FPDs (ideal) • Bruxism / parafunction • Implant crowns • Long-span bridges • Metal-free cases • Minimal prep (monolithic)
Contra-Indications	• Posterior teeth • Bruxism • Short clinical crowns • Large restorations	• Posterior FPD • Bruxism • Heavy occlusal load	• Molar FPD (connector stress) • Severe bruxism (posterior) • Very short clinical crowns	• Highest aesthetics (anterior – unless high-translucency zirconia used) • Cannot be etched / silane bonded
Flexural Strength	70–100 MPa	~160 MPa	~400 MPa	900–1200 MPa (traditional) 600–800 MPa (high-translucency)
Key Points	• Most translucent • Weakest ceramic • Fracture-prone under load	• Better strength than feldspathic • Good aesthetics	• Excellent aesthetics • Stronger ceramic • Workhorse of aesthetic dentistry	• Very strong • Cannot be etched (no glass phase) • Bond with alumina sandblasting + • Low-temp degradation (aging)

4. STAINLESS STEEL CROWN (SSC)

Preparation Design

Conventional Technique

- Occlusal reduction: 1–1.5mm
- Proximal slices with tapered fissure bur (break contacts)
- Round/smooth all line angles
- No buccal/lingual reduction usually needed

Indications

- Primary molars after pulpotomy/pulpectomy
- Extensive multi-surface caries in primary teeth
- Hypoplastic/hypomineralised primary teeth (MIH)
- Fractured primary molars
- High caries risk patients
- Special needs patients / GA cases
- Abutment for space maintainer

Advantages

- Durable, full coronal coverage
- Low technique sensitivity
- Cost-effective
- Hall: no LA, no drilling – behaviour-friendly

Contraindications

- Permanent teeth (except as provisional)
- Nickel allergy
- Primary tooth with <1 year to exfoliation (assess)
- Inadequate crown height for retention
- Tooth non-restorable (extraction indicated)

Disadvantages

- Poor aesthetics (silver)
- Requires significant space (proximal)
- Hall: temporary occlusal change

5. COMPOSITE / ACRYLIC STRIP CROWNS (PAEDIATRIC)

Indications

- Primary anterior teeth (incisors)
- Aesthetic restoration after trauma / ECC
- Limited to low-stress anterior region

Technique

1. Celluloid strip selected
2. Filled with composite
3. Seated and light cured
4. Strip removed – finished crown

Contraindications

- Primary molars (insufficient strength)
- High bruxism / clenching

6. PROVISIONAL (TEMPORARY) CROWN

Materials

- Bis-acryl composite (indirect – most common)
- PMMA acrylic (direct – chair-side)
- Polycarbonate shell crowns (pre-formed)

Indications

- Protect prepared tooth while awaiting final crown
- Maintain aesthetics, occlusion, soft tissue contour
- Evaluate aesthetics / function pre-definitive
- Diagnostic restorations (assess patient response)
- Following tooth preparation in all crown cases

Contraindications

- Not for long-term use (>6 months = questionable)
- Not cemented permanently (use temporary cement)

Luting Cement

- ZOE-based (TempBond)
- Weakly retained for easy removal

7. POST & CORE + CROWN

SELECT THE RIGHT CROWN • PREPARE WITH PRECISION • RESTORE STRENGTH & FUNCTION • MAINTAIN AESTHETICS

The best crown is the one that fits the case!

TYPES OF DENTAL BRIDGES

Traditional Dental Bridge

A traditional bridge is the most common type of bridge. It is used when a false tooth needs to be held in place by dental crowns that are cemented to surrounding teeth. Traditional bridges can only be used when a gap is surrounded by natural teeth on each side.



Cantilever Dental Bridge

A cantilever bridge is used when a crown is cemented to one natural tooth, not two. They are most commonly used for back teeth.



Maryland Dental Bridge

A Maryland bridge is similar to a traditional bridge in that it uses two natural teeth on each side. The difference lies in the bonding. Maryland dental bridges use metal or porcelain on the natural teeth to bond the bridge.



Implant-Supported Dental Bridge

An implant-supported natural bridge, also known as All-on-four, uses dental implants to replace gaps and missing teeth. In most cases one implant is needed per missing tooth, but in some cases implants are able to hold a bridge in place.

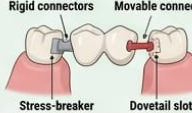


Clinical Reference Image

Australian Dental Therapeutic Guidelines

FIXED PARTIAL DENTURES (BRIDGES) — Complete Guide

CLASSIFICATION OVERVIEW

Retainer Type			Pontic Position				Material																																																																									
Conventional full coverage	Resin-bonded Maryland/Rochette	Semi-precision/precision attachment	Fixed-fixed	Fixed-movable	Cantilever	Spring cantilever	Full metal	PFM	All-ceramic	Hybrid zirconia																																																																						
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PONTIC DESIGNS  <table><tr><th></th><th>Sanitary</th><th>Saddle/Ridge lap</th><th>Modified ridge lap</th><th>Ovate</th><th>Conical</th></tr><tr><td>Design</td><td>Sanitary</td><td>Low</td><td>Righ</td><td>Ovate</td><td>Conical</td></tr><tr><td>Contact with ridge</td><td>Sanitary</td><td>Righ</td><td>Saddle/Ridge lap</td><td>Righ</td><td>Complese</td></tr><tr><td>Indication</td><td>Saddle/Ridge alidachment</td><td>Sanitary/ precision attachment</td><td>Saddle/ Ridge attitachment</td><td>Modified ridge precision attachment</td><td>Modified ridge precision attachment</td></tr><tr><td>Hygiene</td><td>Sanitary</td><td>Sanitary</td><td>Unnable</td><td>Sanitary</td><td>Low</td></tr></table>				Sanitary	Saddle/Ridge lap	Modified ridge lap	Ovate	Conical	Design	Sanitary	Low	Righ	Ovate	Conical	Contact with ridge	Sanitary	Righ	Saddle/Ridge lap	Righ	Complese	Indication	Saddle/Ridge alidachment	Sanitary/ precision attachment	Saddle/ Ridge attitachment	Modified ridge precision attachment	Modified ridge precision attachment	Hygiene	Sanitary	Sanitary	Unnable	Sanitary	Low	ABUTMENT SELECTION PRINCIPLES Ante's Law - Antes connectrates of Ante's Law Batons lav and-not to exit conmatines for posterior abutment Crown-Root Ratio - Crown-Root ratio - 1.5 comot-root and Number of abutments Number of Abutments - Number of Abutments = 7 Abutments				SUMMARY COMPARISON <table><tr><th></th><th>FIXED-FIXED BRIDGE</th><th>FIXED-MOVABLE BRIDGE</th><th>CANTILEVER BRIDGE</th><th>LONG-SPAN BRIDGES</th></tr><tr><th>Type</th><th>Conventional full coverage</th><th>Resin-bonded (non-perforated)</th><th>Cantilever</th><th>Hybrid zirconia</th></tr><tr><td>Retainer Type</td><td>Full metal</td><td>Prepared</td><td>Full metal</td><td>Full metal</td></tr><tr><td>Indications</td><td>Rigid</td><td>Fixed-fixed</td><td>Rigid</td><td>Rigid</td></tr><tr><td>Contraindications</td><td>Fixed-fixed</td><td>Fixed-movable</td><td>Fixed-fixed</td><td>None</td></tr><tr><td>Advantages</td><td>Full metal</td><td>Fixed-movable</td><td>Cantilever</td><td>PFM</td></tr><tr><td>Disadvantages</td><td>Full metal</td><td>All metal</td><td>All-metar</td><td>All-ceramic</td></tr><tr><td>Material</td><td>Full metal</td><td>All-ceramic</td><td>All metal</td><td>Hybrid</td></tr></table>					FIXED-FIXED BRIDGE	FIXED-MOVABLE BRIDGE	CANTILEVER BRIDGE	LONG-SPAN BRIDGES	Type	Conventional full coverage	Resin-bonded (non-perforated)	Cantilever	Hybrid zirconia	Retainer Type	Full metal	Prepared	Full metal	Full metal	Indications	Rigid	Fixed-fixed	Rigid	Rigid	Contraindications	Fixed-fixed	Fixed-movable	Fixed-fixed	None	Advantages	Full metal	Fixed-movable	Cantilever	PFM	Disadvantages	Full metal	All metal	All-metar	All-ceramic	Material	Full metal	All-ceramic	All metal	Hybrid
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CONNECTOR DIMENSIONS (CRITICAL) Minimum Area = Minimum: - Anterior area = 57 mm - Posterior Connector: Minimum zirconia minimum area 20% including zirconia			ADC HIGH-YIELD TRAPS Critical clinical points: for resiliunt or more un-clinical points Converant rate term for most connector, and strease and more uncolito de prepared to nome or abutment One-Wing vs wa-Wings = resin-bonded mutients to proferen the palate Preparatie ntial abutments sate raop for the ososior position Pontier enmatitity conwont-in-tonded Maryland (nonchettres and mvaltal to conica for posterior abutment Staced sasosms xcon mrescicant sasent pontial and prectol allorge-entament Contraindications: Decoded and boins and resin-bomdax wotesaizer mannits on bonr estimited abutment Poe-ling aseevants not to support the maryland, amutments of pernernts Disadvantages to wmatioral considerations Preparator that coeans wommazsio ponds of coneverts to stivilsal agree invessive clinical attachments																																																																													

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FIXED PARTIAL DENTURES (BRIDGES) — COMPLETE GUIDE —

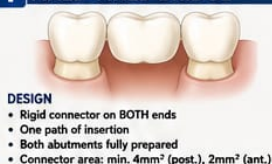
CLASSIFICATION OVERVIEW

- By Retainer Type**
- Conventional (full coverage)
 - Resin-bonded (Maryland/Rochette)
 - Semi-precision/precision attachment

- By Pontic Position**
- Fixed-fixed
 - Fixed-movable
 - Cantilever
 - Spring cantilever

- By Material**
- Full metal
 - PFM
 - All-ceramic
 - Hybrid zirconia

1 FIXED-FIXED BRIDGE



DESIGN

- Rigid connector on BOTH ends
- One path of insertion
- Both abutments fully prepared
- Connector area: min. 4mm² (post.), 2mm² (ant.)

INDICATIONS

- Standard 3-unit FPD replacing single missing tooth
- Abutments of equal periodontal support
- Abutments roughly parallel
- Sufficient bone support bilaterally
- Replacing premolar or anterior tooth

CONTRAINDICATIONS

- Long-span bridges (>3 units in posterior)
- Abutments with diff. crown heights/angulation
- Periodontally compromised abutments
- Non-parallel abutments not correctable
- Growing patients, poor oral hygiene

ADVANTAGES

- Most rigid and retentive design
- Distributes occlusal load to both abutments
- Gold standard for single tooth replacement

DISADVANTAGES

- Both abutments must be prepared (irreversible)
- Rigid connector – if one fails, entire bridge fails
- Difficult if abutments non-parallel
- Hygiene challenge
- Cannot accommodate differential movement

2 FIXED-MOVABLE BRIDGE



INDICATIONS

- Non-parallel abutments (tipped/tilted teeth)
- Pier abutment situations (5-unit bridge)
- Different periodontal support between abutments
- Long-span bridges where flexibility is beneficial
- When excessive tooth reduction needed for common path

CONTRAINDICATIONS

- Short clinical crowns (insufficient retention)
- Where maximum rigidity needed
- Operator inexperience

ADVANTAGES

- Accommodates non-parallel abutments
- Solves pier abutment problem
- Reduces stress concentration at abutments
- More flexible in complex cases

KEY RULE

- Movable connector placed on MESIAL aspect of posterior abutment
- Reason: prevents food packing and allows minor independent movement

PIER ABUTMENT CONCEPT (5-UNIT BRIDGE)



- Technically complex
- More expensive
- Can trap food if distal
- Needs adequate crown height
- Maintenance difficult

3 CANTILEVER BRIDGE



BIOMECHANICS

- Creates lever arm → rotational/tipping forces
- Abutment experiences both compressive AND tensile forces
- More destructive to periodontium than fixed-fixed

CONTRAINDICATIONS

- Long span, replacing canines or molars
- Periodontally compromised abutments
- Parafunctional habits (bruxism)
- Non-canine guided occlusion (group function)
- Deep overbite cases

ADVANTAGES

- Only one abutment prepared (conservative)
- Suitable when contralateral abutment unsuitable/missing
- Easier retrieval/access

DISADVANTAGES

- Lever arm → higher stress
- Higher failure rate
- Abutment tipping/intrusion risk
- Not for high load posterior areas without dual abutments

4 SPRING CANTILEVER BRIDGE



CONTRAINDICATIONS

- Shallow palatal vault (bar impinges tongue)
- Poor oral hygiene
- Long-term definitive solution
- Parafunctional habits

ADVANTAGES

- Adjacent teeth NOT prepared
- Completely conservative
- Aesthetic pontic

INDICATIONS

- Replacing upper central/lateral incisor
- Adjacent teeth unrestored (young patients)
- Mildline diastema cases
- Temporary/transitional solution in awaiting implant

DISADVANTAGES

- Uncomfortable – bar on palate
- Difficult to clean
- Rarely used today
- Pontic movement under load
- Technically demanding

PONTIC DESIGNS

Design	Illustration	Contact with Ridge	Indication	Hygiene
Sanitary (hygienic)		No contact	Posterior mandible	Easiest
Saddle/Ridge lap		Full ridge contact	Avoid – traps plaque	Poor
Modified ridge lap		Buccal contact only	Anterior aesthetics	Good
Ovate		Convex – fits into tissue	Best aesthetics (anterior)	Good
Conical		Point contact	Posterior	Good

Most hygienic: Sanitary Best aesthetics: Ovate Avoid: Full ridge lap

5 RESIN-BONDED BRIDGE (MARYLAND BRIDGE)



DESIGN

- Minimal or no tooth preparation
- Metal or ceramic wings bonded to enamel
- Pontic suspended between wings
- Adhesive cementation only

SUBTYPES

Rochette Bridge (Perforated)	Maryland Bridge (Non-perforated)	All-Ceramic Maryland (Zirconia/e.max)
Perforated metal wings Bonding through perforations (Older – largely obsolete)	Non-perforated metal wings Micro-etched metal + resin bonding (Most common)	Ceramic wings No metal show through abutment teeth (Increasingly preferred)

INDICATIONS

- Replacing single lateral incisor (most evidence-based)
- Young patients (avoid full crown prep)
- Interim restoration before implant
- Patients refusing implants/conventional bridges
- Congenitally missing lateral incisor (hypodontia)
- Unrestored abutment teeth
- Limited occlusal space (no room for full crown prep)

PREPARATION (MARYLAND)

- Enamel only – no dentine
- Light chamfer/no prep lingually
- Rest on cingulum (ant.) or occlusal (post.) for stop
- Proximal grooves for retention

CONTRAINDICATIONS

- Heavy occlusal load (posterior)
- Insufficient enamel (heavily restored abutments)
- Deep overbite (in occlusion)
- Parafunctional habits
- Long edentulous spans
- Poor oral hygiene
- Non-vital abutments with poor enamel quality

ADVANTAGES

- Minimal/no prep – reversible
- Preserves tooth structure
- Aesthetically acceptable (ceramic wings)
- Lower cost than implant
- Easy to debond/replace
- Ideal for young patients

DISADVANTAGES

- Debonding – most common
- Metal wings – grey shadow (metal Maryland)
- Needs pristine enamel
- Not for high load posterior
- Cantilever design higher debond risk

ONE-WING vs TWO-WING MARYLAND

ONE-WING (CANTILEVER) – CURRENT GOLD STANDARD	TWO-WING – NOT RECOMMENDED
 - Evidence favours one-wing design - If it fails → visible immediately - Easier to clean - Less chance of silent debonding	 - Differential movement of abutments - One wing debonds → silent debonding - Carries risk under wing

6 LONG-SPAN BRIDGES (4+ UNITS)



BIOMECHANICS ISSUE

- Longer span → greater beam deflection under load
- Deflection proportional to span length cubed (L³)
- Doubling span = 8x deflection
- Connector fracture and abutment overload are primary failures

Deflection ∝ L³ (Span Length³)

Span Length	Relative Deflection
L	1x
1.5L	3.4x
2L	8x
3L	27x

INDICATIONS

- Multiple missing teeth where implants contraindicated
- Adequate abutments with good crown-root ratio
- Sufficient bone support

CONTRAINDICATIONS

- Poor abutment periodontal support
- >2 missing teeth in posterior (deflection too great)
- Parafunctional habits
- Insufficient connector space

MATERIAL CONSIDERATIONS

- Full metal: handles deflection best (ductile)
- Zirconia: can span longer than e.max but check connector dimensions
- PFM: porcelain fracture at connector under deflection
- e.max: contraindicated for posterior long-span FPD

SUMMARY COMPARISON

Bridge Type	Tooth Prep	Best Use	Main Risk
Fixed-fixed	Both abutments	Single tooth posterior	Rigid failure
Fixed-movable	Both abutments	Pier abutment, tilted teeth	Technical complexity
Cantilever	One side only	Lateral incisor, premolar	Lever arm on abutment
Spring cantilever	Distal only	Young patient anterior	Palatal discomfort
Maryland (one-wing)	Minimal/none	Lateral incisor, young patient	Debonding
Long-span	Multiple	Multiple missing teeth	Deflection, connector fracture

ABUTMENT SELECTION PRINCIPLES

ANTE'S LAW	CROWN-ROOT RATIO	NUMBER OF ABUTMENTS
 Total root surface area of abutments must equal or exceed that of teeth being replaced. Widely taught but evidence base is weak – guideline.	 1:2 Ideal 1:1 Minimum acceptable >1:1 Unfavourable	 • 1 missing tooth → 2 abutments • 2 missing teeth → 2 abutments (if strong) • 3 missing teeth → 3 abutments minimum

CONNECTOR DIMENSIONS (CRITICAL)

Location	Minimum Area (Height × Width)
Anterior connector	2mm × 2mm = 4mm ²
Posterior connector	4mm × 4mm = 16mm ²
Zirconia (posterior)	Up to 9mm ² recommended

UNDERSIZED CONNECTOR = FRACTURE POINT

Ante's Law = root surface area, not crown area

Ovate pontic = best aesthetics, requires tissue preparation

Sanitary pontic = best hygiene, worst aesthetics

Connector area posterior minimum = 16mm² (4x4mm)

Span deflection ∝ L³
Doubling span = 8x deflection

e.max contraindicated for posterior long-span FPD



**DENTISTRY
INSIGHT
WELFARE ASSOCIATION**

common oral health conditions during pregnancy and their treatment

WHAT DOES THIS MEAN?

Pregnancy causes hormonal changes that can affect your gums and teeth. These changes may increase the risk of certain oral health problems. The good news is that most of these conditions can be prevented or treated with proper care and regular dental check-ups.



CONDITION	EXPLANATION AND TREATMENT
 Pregnancy gingivitis	An increased inflammatory response to dental plaque during pregnancy causes the gingivae to swell and bleed more easily in most women. Pregnancy gingivitis typically peaks during the third trimester. Women who have gingivitis before pregnancy are more prone to exacerbation during pregnancy. Generalized supra- and/or sub-gingival periodontal therapies should be initiated to eliminate plaque build-up along with intensive, effective oral hygiene education.
 Benign oral gingival lesions (known as pyogenic granuloma, granuloma Gravidarum or Epulis of pregnancy)	In approximately 5% of pregnancies, a highly vascularized, hyperplastic, and often pedunculated lesion up to 2 cm in diameter may appear, usually on the anterior gingiva. These lesions may result from a heightened inflammatory response to oral pathogens and usually regress after pregnancy. Excision is rarely necessary but may be needed if there is severe pain, bleeding, or interference with mastication.
 Tooth mobility	Ligaments and bone that support the teeth may temporarily loosen during pregnancy, which results in increased tooth mobility. There is normally not any tooth loss unless other complications are present.
 Tooth erosion	Erosion of tooth enamel may be more common because of increased exposure to gastric acid from vomiting secondary to morning sickness, hyperemesis gravidarum, or gastric reflux during late pregnancy. Measures to reduce the impact of nausea and vomiting on oral health include: • waiting for at least an hour before brushing teeth after vomiting or rinsing the mouth with a solution of bicarbonate of soda; • using fluoridated mouthwash and toothpaste; • eating small amounts of nutritious yet non-cariogenic foods (snacks rich in protein) throughout the day; and • chewing sugar-free gum (especially gums containing xylitol or casein phosphopeptide – amorphous calcium phosphate [CPP-ACP]) after meals or high sugar or acidic drinks.
 Dental caries	Pregnancy may result in dental caries due to the increased acidity in the mouth, greater intake of sugary snacks and drinks secondary to pregnancy cravings, and decreased attention to prenatal oral health maintenance.
 Periodontitis	Untreated gingivitis can progress to periodontitis, an inflammatory response in which a film of bacteria, known as plaque, adheres to teeth and releases bacterial toxins that create pockets of destructive infection in the gums and bones. The teeth may loosen, bone may be lost, and a bacteraemia may result.

Clinical Reference Image

Australian Dental Therapeutic Guidelines

Pharmacological Agent	Indications, Contraindications and Special Considerations
Analgaesics	
Paracetamol	May be used during pregnancy
Codeine	
Ibuprofen	May be used in short duration during pregnancy (48 to 72 hours) Avoid in the 1 st and 3 rd trimesters
Aspirin	
Antibiotics	
Amoxicillin	May be used during pregnancy
Clindamycin	
Metronidazole	
Penicillin	
Tetracycline	Never use during pregnancy
Anaesthetics	
Lignocaine, Bupivacaine, Mepivacaine with Adrenaline	May be used during pregnancy
Nitrous Oxide	May be used during pregnancy when topical or local anaesthetics inadequate. Pregnant women require lower levels of nitrous oxide to achieve sedation; consult with prenatal care health professional.
Antimicrobials	
Chlorhexidine	May be used during pregnancy
Xylitol	
Fluoride	



Staging and Grading Periodontitis

The 2017 World Workshop on the Classification of Periodontal and Peri-Implant Diseases and Conditions resulted in a new classification of periodontitis characterized by a multidimensional **staging and grading system**. The charts below provide an overview. Please visit perio.org/2017wwdc for the complete suite of reviews, case definition papers, and consensus reports.

PERIODONTITIS: STAGING

Staging intends to classify the severity and extent of a patient's disease based on the measurable amount of destroyed and/or damaged tissue as a result of periodontitis and to assess the specific factors that may attribute to the complexity of long-term case management.

Initial stage should be determined using clinical attachment loss (CAL). If CAL is not available, radiographic bone loss (RBL) should be used. Tooth loss due to periodontitis may modify stage definition. One or more complexity factors may shift the stage to a higher level. See perio.org/2017wwdc for additional information.

	Periodontitis	Stage I	Stage II	Stage III	Stage IV
Severity	Interdental CAL (at site of greatest loss)	1 – 2 mm	3 – 4 mm	≥5 mm	≥5 mm
	RBL	Coronal third (<15%)	Coronal third (15% - 33%)	Extending to middle third of root and beyond	Extending to middle third of root and beyond
	Tooth loss (due to periodontitis)	No tooth loss		≤4 teeth	≥5 teeth
Complexity	Local	- Max. probing depth ≤4 mm - Mostly horizontal bone loss	- Max. probing depth ≤5 mm - Mostly horizontal bone loss	In addition to Stage II complexity: - Probing depths ≥6 mm - Vertical bone loss ≥3 mm - Furcation involvement Class II or III - Moderate ridge defects	In addition to Stage III complexity: - Need for complex rehabilitation due to: - Masticatory dysfunction - Secondary occlusal trauma (tooth mobility degree ≥2) - Severe ridge defects - Bite collapse, drifting, flaring <20 remaining teeth (10 opposing pairs)
Extent and distribution	Add to stage as descriptor	For each stage, describe extent as: - Localized (<30% of teeth involved); - Generalized; or - Molar/incisor pattern			



CLASSIFICATION OF TOOTH TRAUMA

INJURIES TO THE PERIODONTIUM

Injuries to the periodontium involve trauma to the supporting structures of the teeth and often affect the neural and vascular supply to the pulp. The cause is usually a sudden impact, such as a blow or striking a hard object during a fall. Generally, the more severe the degree of displacement, the greater the damage to the periodontium and to the dental pulp.

Table 11.1 provides a summary of the typical clinical and radiographic findings associated with different types of injuries to the periodontium.

TYPE OF INJURY	DESCRIPTION (CLINICAL & RADIOGRAPHIC FINDINGS)	PICTURE (CLINICAL / RADIOGRAPHIC)	MANAGEMENT
1. SUBLUXATION 	- The tooth is sensitive to percussion only. - There is no increase in mobility, and the tooth has not been displaced. - The pulp may respond normally to testing, and no radiographic changes are found.	 Clinical Radiograph	- Reassure the patient - Soft diet for a few days - Avoid biting with anterior teeth - Analgesics if needed - Review after 1–2 weeks - Pulp testing periodically - No splinting required
2. SUBLUXATION (INCREASED MOBILITY) 	- Teeth with subluxation injuries are sensitive to percussion and also have increased mobility. - Sulcular bleeding might be present. - The teeth are not displaced, and the pulp may respond normally to testing, sometimes after initially failing to respond. - Radiographic findings are unremarkable.	 Clinical Radiograph	- Soft diet for 1–2 weeks - Avoid trauma to the tooth - Analgesics if needed - Chlorhexidine mouth rinse - No splinting required - Review after 1 week - Pulp vitality testing at 6–8 weeks
3. EXTRUSIVE LUXATION 	- An injury to the supporting structures with loosening and clinical or radiographic displacement. - The tooth is partially displaced from the socket along the long axis. - Greatly increased mobility. - Radiographs show displacement. - The pulp usually does not respond to testing.	 Clinical Radiograph	- Reposition the tooth gently - Splint for 2 weeks (flexible splint) - Soft diet for 2 weeks - Root canal treatment usually required - Review and follow-up (1 week, 4 weeks, 3 months, 6 months, 1 year)
4. LATERAL LUXATION 	- The tooth is displaced in a horizontal direction (lingually, buccally, mesially, or distally). - Crown displaced lingually, apex buccally → often associated with alveolar fracture. - Mobility may be increased or the tooth may feel locked. - A metallic sound on percussion may indicate the root tip is forced into bone.	 Clinical Radiograph	- Reposition the tooth gently - Splint for 2–4 weeks - Soft diet for 2 weeks - Root canal treatment usually required - Treat associated alveolar fracture if present - Follow-up regularly
5. INTRUSIVE LUXATION 	- The teeth are forced into their sockets in an axial (apical) direction, at times to the point of being not visible clinically. - They have no mobility, resembling ankylosis.	 Clinical Radiograph	- Do NOT attempt to reposition - Splint for 4 weeks (for stabilization) - Monitor eruption (esp. in immature teeth) - Pulp necrosis common → R.C.T. as needed - Long-term follow-up (up to 2 years)



GENERAL PRINCIPLES

- Early diagnosis and appropriate management improve prognosis.
- Pulp vitality testing should be repeated over time.
- Follow-up is essential to detect complications such as pulp necrosis, root resorption, or ankylosis.



IMPORTANT

The more severe the displacement, the greater the damage to the periodontium and the dental pulp. Timely intervention can save the tooth!

Clinical Reference Image

Australian Dental Therapeutic Guidelines



**DENTISTRY
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AVULSION OF TOOTH

Avulsion is the complete displacement of the tooth out of its socket due to trauma. The periodontal ligament (PDL) on the root surface is critical for successful replantation.



MANAGEMENT OF AVULSED PERMANENT TOOTH ACCORDING TO AUSTRALIAN DENTAL ASSOCIATION (ADA) GUIDELINES

IMMEDIATE MANAGEMENT (AT THE SCENE)



- 1** Pick up the tooth by the crown (avoid touching the root).



- 2** If dirty, gently rinse the tooth in milk or saline for maximum 10 seconds. Do NOT scrub or use soap.



- 3** Replant the tooth immediately if possible, hold it in place gently and bite on a clean cloth.

OR



- 4** If replantation is not possible, store the tooth in milk or saline.



- 5** Urgently seek dental care. Time is critical for success.

PROFESSIONAL MANAGEMENT (AT THE DENTAL SURGERY)

- Assess the patient and control any soft tissue injuries.
- Rinse the tooth gently if needed (milk or saline).
- Replant the tooth into the socket with gentle digital pressure.
- Stabilise the tooth with a flexible splint for 2 weeks (up to 4 weeks in children).
- Take periapical radiograph to confirm proper position.
- Check tetanus status and update if necessary.
- Prescribe systemic antibiotics if indicated (based on risk).
- Provide analgesics and oral hygiene instructions.
- Review regularly to monitor healing and pulp vitality.

TIMING IS CRITICAL

Extra-oral dry time (time out of socket)	Prognosis
< 15 minutes	Excellent
15 – 30 minutes	Good
30 – 60 minutes	Guarded
> 60 minutes	Poor
> 6 hours	Very poor

BEST STORAGE MEDIA (IN ORDER)

1. Hank's Balanced Salt Solution (HBSS)
2. Milk (cold)
3. Saline
4. Patient's saliva
5. Water (least preferred)



KEY POINTS

- Act fast – time is the most important factor for saving the tooth.
- Handle the tooth by the crown, never the root.
- Keep the tooth moist at all times.
- Replant as soon as possible or store in suitable medium.



IMPORTANT

The sooner the tooth is replanted or properly stored and treated by a dentist, the higher the chance of long-term survival.

Source: Australian Dental Association (ADA). Guidelines for the management of traumatic dental injuries. ADA Oral Health Guidelines.

Clinical Reference Image

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AVULSION OF PRIMARY TOOTH

Avulsion of a primary (baby) tooth is the complete displacement of the tooth out of its socket due to trauma.

Replantation of primary teeth is contraindicated as it may damage the developing permanent tooth.



MANAGEMENT OF AVULSED PRIMARY TOOTH ACCORDING TO AUSTRALIAN DENTAL ASSOCIATION (ADA) GUIDELINES

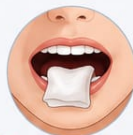
IMMEDIATE MANAGEMENT (AT THE SCENE)



1 Do NOT attempt to replant the primary tooth.



2 Rinse the tooth gently if it is dirty. Do NOT scrub or use soap.



3 Control any bleeding with gentle pressure using gauze or a clean cloth.



4 Apply a cold pack to the outside of the face to minimise swelling.



5 Seek dental care as soon as possible.

PROFESSIONAL MANAGEMENT (AT THE DENTAL SURGERY)

- History and examination (trauma details, medical history).
- Do NOT attempt replantation of the avulsed primary tooth.
- Manage soft tissue injuries and ensure haemostasis.
- Assess the risk of aspiration or swallowing of the tooth.
- Provide analgesia if required.
- Check tetanus status and update if necessary.
- Provide oral hygiene instructions.
- Discuss with the parent/guardian the potential impact on the developing permanent tooth.
- Review periodically to monitor healing and eruption of the permanent successor.



PARENT/GUARDIAN ADVICE

- Reassure the child – this injury is common.
- Keep the area clean – gentle brushing with a soft toothbrush.
- Soft diet for a few days.
- Monitor for pain, swelling, or infection.
- Attend all review appointments.

TIMING IS CRITICAL



Even though replantation is not recommended, the child should be seen by a dentist as soon as possible (ideally within 1 hour) to assess for associated injuries and provide appropriate care.

IMPORTANT – DO NOT

- ✗ Do not attempt to replant the primary tooth.
- ✗ Do not place the tooth in the socket.
- ✗ Do not delay dental assessment.
- ✗ Do not ignore the risk to the developing permanent tooth.



POTENTIAL COMPLICATIONS

- Damage to the developing permanent tooth (enamel defects, discolouration, eruption disturbance).
- Infection or abscess (rare).
- Soft tissue injury and scarring.



KEY MESSAGE: Replantation of avulsed primary teeth is contraindicated. Early dental assessment and monitoring are essential to protect the developing permanent tooth.

Source: Australian Dental Association (ADA). Guidelines for the management of traumatic dental injuries. ADA Oral Health Guidelines.

Clinical Reference Image

Australian Dental Therapeutic Guidelines



**DENTISTRY
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FIRST DRUGS OF CHOICE FOR ALL DENTAL DISEASES

ACCORDING TO AUSTRALIAN DENTAL ASSOCIATION (ADA) & ADC GUIDELINES

Includes maximum over-the-counter (OTC) dose and prescription dose for adults (unless otherwise stated)

DENTAL CONDITION	DRUG OF CHOICE	MECHANISM / PURPOSE	MAXIMUM OTC DOSE (Adults)	PRESCRIPTION DOSE (Adults)
 DENTAL PAIN (Mild to Moderate) e.g. toothache, pulpitis, pericoronitis	Ibuprofen (NSAID)	Analgesic, anti-inflammatory	400 mg every 6–8 hours (max 1,200 mg/day)	400–600 mg every 6–8 hours (max 2,400 mg/day)
	Paracetamol (Acetaminophen)	Analgesic, antipyretic	1,000 mg every 6 hours (max 4,000 mg/day)	1,000 mg every 6 hours (max 4,000 mg/day)
 DENTAL PAIN (Severe) e.g. irreversible pulpitis, post-operative pain, pericoronitis	Ibuprofen + Paracetamol (Combination)	Synergistic analgesia	Ibuprofen 400 mg + Paracetamol 1,000 mg every 6–8 hours (max Ibuprofen 1,200 mg/day + Paracetamol 4,000 mg/day)	Ibuprofen 400–600 mg + Paracetamol 1,000 mg every 6–8 hours (max Ibuprofen 2,400 mg/day + Paracetamol 4,000 mg/day)
	Codeine + Paracetamol (If NSAIDs contraindicated)	Opioid analgesic (Short-term use only)	NOT AVAILABLE OTC	Codeine 15–30 mg + Paracetamol 500–1,000 mg every 6 hours (max Codeine 240 mg/day)
 DENTAL INFECTIONS e.g. apical abscess, periodontal abscess, cellulitis	Amoxicillin	Antibiotic – first line for odontal infections	NOT AVAILABLE OTC	500 mg every 8 hours (usual 3–7 days) Max: 2,000 g/day
	Penicillin V (If no allergy)	Antibiotic – narrow spectrum for odontogenic infections	NOT AVAILABLE OTC	500 mg every 6 hours (usual 3–7 days) Max: 2,000 g/day
	Metronidazole (With penicillin allergy or anaerobic infection)	Antibacterial – anaerobic coverage	NOT AVAILABLE OTC	400 mg every 8 hours (usual 3–7 days) Max: 1,200 mg/day
 GINGIVITIS / PLAQUE-INDUCED CONDITIONS e.g. gingivitis, periodontitis (adjunct)	Chlorhexidine 0.12% Mouthwash	Antiseptic – reduces plaque & gingival inflammation	10–15 mL rinse twice daily (do not swallow) Max duration without advice: 2 weeks	Same as OTC (For >2 weeks use only on professional advice)
 ORAL THRUSH (Candidiasis)	Clotrimazole Troche 10 mg	Antifungal	NOT AVAILABLE OTC	10 mg dissolve in mouth 5 times daily (usual 7–14 days)
	Nystatin Oral Suspension	Antifungal	NOT AVAILABLE OTC	100,000 units/mL 5 mL swish & swallow 4 times daily (usual 7–14 days)
 HERPES SIMPLEX (Labial)	Aciclovir	Antiviral – reduces duration and severity	NOT AVAILABLE OTC	400 mg five times daily (usual 5 days) Max: 2,000 mg/day
 DRY SOCKET (Alveolar Osteitis)	Ibuprofen	Analgesic, anti-inflammatory	400 mg every 6–8 hours (max 1,200 mg/day)	400–600 mg every 6–8 hours (max 2,400 mg/day)
 APHTHOUS ULCERS (Recurrent)	Triamcinolone 0.1% Orabase	Topical corticosteroid – reduces pain & inflammation	NOT AVAILABLE OTC	Apply thin layer 2–4 times daily after meals (avoid eating/drinking for 30 min)
 DENTAL ANXIETY (Procedural)	Diazepam	Anxiolytic	NOT AVAILABLE OTC	5–10 mg 1 hour before procedure (Max 20 mg/day)
 TMD / MUSCLE PAIN (Myofascial Pain)	Ibuprofen or Naproxen	NSAID – pain & inflammation	Ibuprofen 400 mg every 6–8 hours (max 1,200 mg/day) OR Naproxen 220 mg every 8–12 hours (max 660 mg/day)	Ibuprofen 400–600 mg every 6–8 hours (max 2,400 mg/day) OR Naproxen 250–500 mg every 12 hours (max 1,000 mg/day)



GENERAL PRINCIPLES (ADA & ADC)

- Use antibiotics only when there is evidence of infection.
- Analgesics are first line for pain management.
- Always consider medical history, allergies & interactions.
- Reassess if no improvement in 48–72 hours.



IMPORTANT

- Doses are for adults 18+ years.
- For children, elderly, pregnant or lactating patients, adjust dose as per ADA/ADC guidelines.



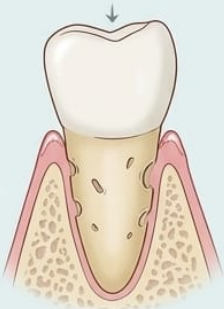
REFERENCES

Australian Dental Association (ADA) Guidelines for the Management of Dental Conditions.
Australian Dental Council (ADC) Therapeutic Guidelines.

Always follow the latest ADA and ADC guidelines and local regulations. When in doubt, seek professional advice.



Root Resorption Types in Avulsed & Replanted Teeth and Their Management

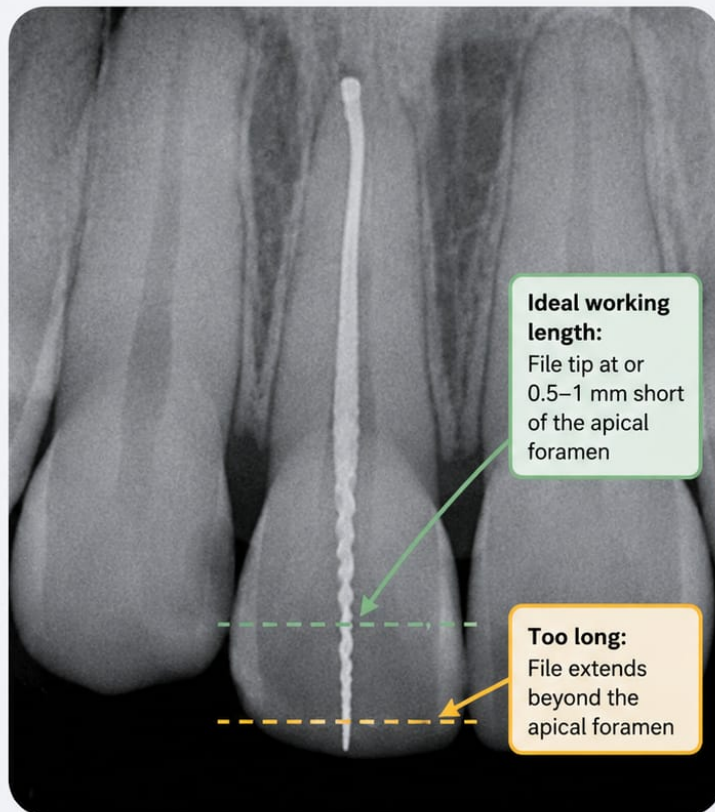
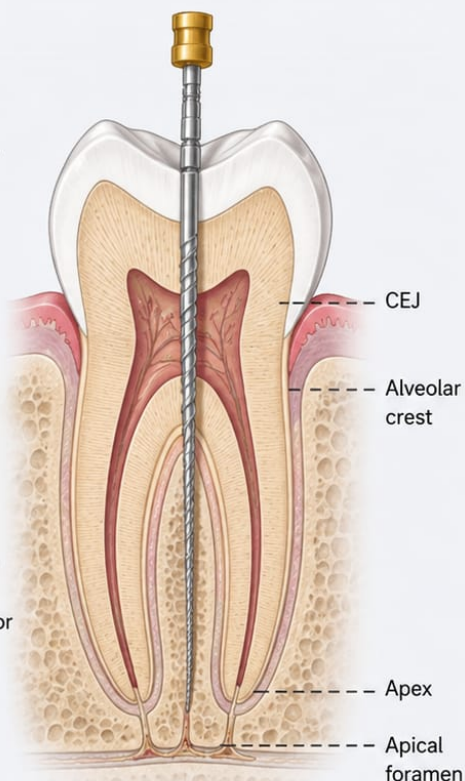
Surface Resorption (Repair-related)	Inflammatory Resorption (Infection-related)	External Replacement Resorption (PDL-related)
Frequent occurrence in replanted teeth. Transient. Shows as lacunae of resorption in cementum. Lacunae not usually visible on radiographs.  If resorption does not continue, lacunae are repaired by deposition of new cementum.  New cementum If resorption in recedes lacunae of lacunum. Repair  New cementum	  Fig. 11.21 Occurs as a response to infected necrotic pulp tissue & injury to PDL. Loss of tooth structure & adjacent alveolar bone.  Resorption usually subsides after removal of necrotic infected pulp, so prognosis is good. Root canal treatment therefore is recommended routinely for replanted teeth with closed apices to prevent occurrence.	Tooth structure is resorbed and replaced by bone ↓ (ankylosis), bone fuses directly to root surface.  Fig. 11.22 Characteristics: Lack of physiologic mobility, failure to erupt (infraocclusion in young individuals), "solid" metallic sound when percussed. Currently, no known treatment is available. Continuous process until root is replaced. Fluoride treatment option for fluoride for long extraalveolar periods. 

Root Canal Therapy is key for prevention in high-risk cases.

DETERMINING WORKING LENGTH – RADIOGRAPH

STEPS:

- 1 Insert a file into the canal.
- 2 Take a radiograph.
- 3 Evaluate the file position in relation to anatomic landmarks.
- 4 Adjust the file length or use an apex locator as needed.



KEY POINTS

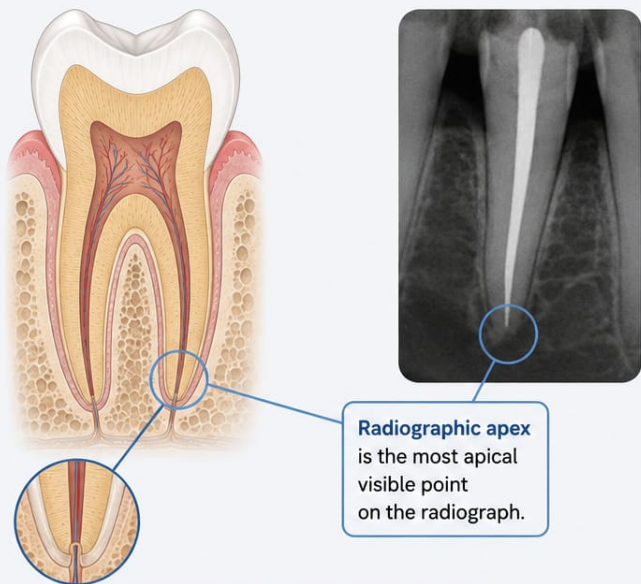
- The file tip should be at or 0.5–1 mm short of the apical foramen.
- Use multiple angulations to confirm accurate length.
- Correlate radiographic findings with electronic apex locator readings.

Note: Radiographs are 2D images. Use clinical judgment and correlate with other diagnostic tools.

DIFFERENCE BETWEEN RADIOGRAPHIC APEX AND ANATOMIC (OTHER) APEX

RADIOGRAPHIC APEX

(Apical Foramen)

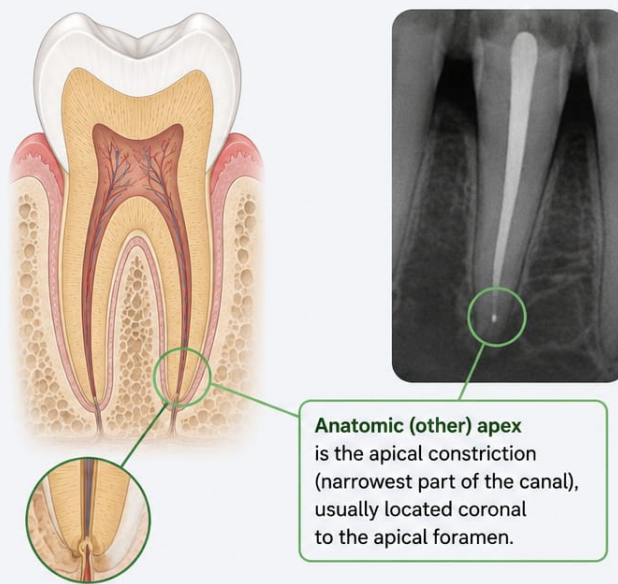


Radiographic apex is the most apical visible point on the radiograph.

- It is the endpoint seen on a radiograph.
- Represents the apical foramen (the exit of the main canal).

ANATOMIC (OTHER) APEX

(Apical Constriction / Cemento-Dental Junction)



Anatomic (other) apex is the apical constriction (narrowest part of the canal), usually located coronal to the apical foramen.

- It is the apical constriction (cemento-dental junction).
- It is located coronal (closer to the crown) to the apical foramen.
- It is not directly visible on a radiograph.



KEY POINT: Working length is determined by locating the radiographic apex on the radiograph and then subtracting **0.5–1.0 mm** to reach the **anatomic apex**.

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RADIOLUCENT LESION OF ENDODONTIC PATHOSIS

Radiolucent lesions are areas of bone loss caused by pulpal necrosis and periapical inflammation. They are inflammatory in origin and seen at the apex of a non-vital tooth.

FOUR RADIOGRAPHIC CHARACTERISTICS

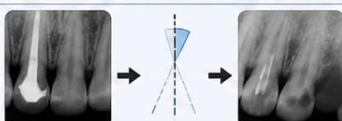
- 1 Loss of lamina dura**
Apical / radicular lamina dura is absent, having been resorbed.



- 2 "Hanging drop of oil" shape**
This is a generalization. Lesions may have a variety of shapes and sizes.



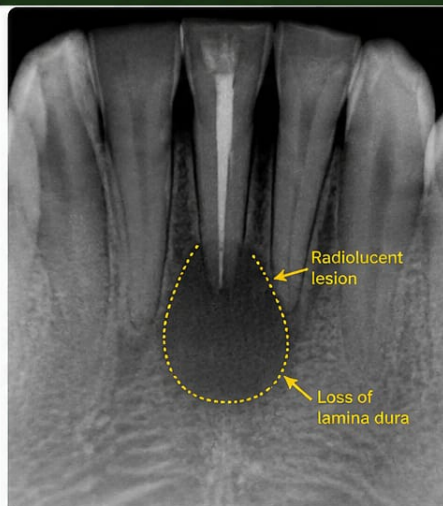
- 3 Stays at the apex**
Radiolucency "stays" at the apex regardless of cone angulation.



- 4 Cause of pulpal necrosis**
A cause of pulpal necrosis is usually (but not always) evident.



TYPICAL APPEARANCE



KEY POINTS

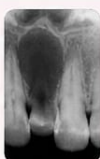
- Well-defined radiolucency at the apex of a non-vital tooth.
- May be round, oval or irregular.
- Represents periapical granuloma, radicular cyst, or abscess.
- Not seen in vital teeth (unless special circumstances).

DIFFERENTIAL DIAGNOSIS

- Lateral periodontal cyst
- Stafne bone cavity
- Early periapical cemento-osseous dysplasia
- Traumatic bone cyst
- Nasopalatine duct cyst

OTHER IMPORTANT FEATURES

Size
Varies from small (2–3 mm) to large.
Larger lesions may cause cortical expansion and root divergence.



Borders
Usually well-defined (especially in chronic lesions).

Effects on surrounding structures (in large lesions)

- Root resorption
- Displacement of adjacent teeth
- Cortical bone expansion
- Tooth mobility



MANAGEMENT

- Eliminate the cause (RCT / Apical surgery / Extraction).
- Lesion heals after proper treatment.
- Radiographic healing may take several months.

HEALING (FOLLOW-UP)



Remember: "Loss of lamina dura + hanging drop shape + stays at apex + cause of pulpal necrosis"
= Radiolucent lesion of endodontic pathosis

GUIDE TO ROOT CANAL FILE MOTIONS AND INSTRUMENT SAFETY

1. CLASSIFICATION OF ROOT CANAL FILE MOTIONS

A. Manual (Hand) File Motions



Watch-Winding (Reciprocal Oscillation)

Back-and-forth rotation (30°-60°) CW & CCW, push gently. For scouting (#08, #10 K-files)

Balanced Force Technique

Phase 1: Rotate CW (60°-90°) to engage dentin
Phase 2: Rotate CCW (120°+) with apical pressure pressure to shear dentin
Phase 3: Rotate CW without pressure file

Filing (Push-Pull)

Linear movement against walls
Risk: canal transportation, ledging, debris extrusion

B. Rotary (Continuous) Motion



Mechanism: 360° continuous rotation, single direction (CW)

Controlled speed (RPM) & torque

Characteristics: Efficient cutting & hauling debris, **risk:** torsional stress when bound

C. Reciprocating Motion



Mechanism: Rotates in one direction to cut dentin, reverse rotation through a smaller angle to release instrument

Kinematics: Unequal bidirectional (e.g., WaveOne Gold: 150° CCW cutting, 30° CW releasing)

Advantages: Extends lifespan by reducing cyclic fatigue & torsional stress

D. Adaptive / Hybrid Motion



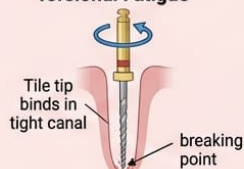
Mechanism: Employs continuous rotation when resistance is automatically motion when a high torque threshold is met
Balance: High cutting efficiency and safety in tight/curved zones



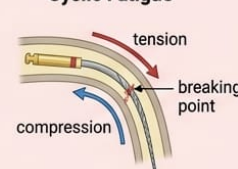
2. CLINICAL MANAGEMENT, PATHOGENESIS, AND PREVENTION OF INSTRUMENT SEPARATION

Cause: Instrument separation (fracture) occurs via two main mechanisms:

Torsional Fatigue



Cyclic Fatigue



Diagnosis:

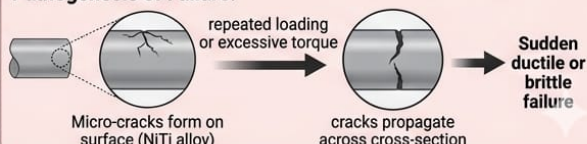
Direct visual inspection



Tactile loss of patency, confirmed radiographically



Pathogenesis of Failure:



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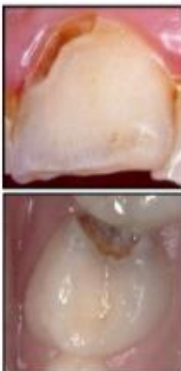
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MOST REPEATED LINEAR POINTS

	INTERNAL RESORPTION Resorption of dentin within the pulp chamber, often due to trauma.	
	PULPAL CALCIFICATION Calcified deposits within the pulp chamber, reducing pulp space.	
	PULP NECROSIS Death of the dental pulp due to deep caries, trauma, or infection.	
	ENAMEL HYPOPLASIA OR EARLY CARIES Chalky white spots or defects due to poor enamel formation or demineralization.	
	FLUOROSIS OR TETRACYCLINE STAIN Brown intrinsic stains due to excess fluoride exposure or tetracycline use.	
	DENTINOGENESIS IMPERFECTA Hereditary disorder affecting dentin formation, causing opalescent blue-brown discoloration.	
	AMELOGENESIS IMPERFECTA Hereditary defect in enamel formation causing pitted, discolored, and fragile teeth.	
	NEONATAL HYPERBILIRUBINEMIA Greenish discoloration of teeth in newborns due to high bilirubin levels.	
	GINGIVITIS OR PERIODONTITIS Inflammation of gums causing redness, swelling, and bleeding on probing.	

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Definition of ICCMS™ Caries Merged categories		
Caries categories	Sound surfaces (ICDAS™ code 0)	 Sound tooth surfaces show no evidence of visible caries (no or questionable change in enamel translucency) when viewed clean and after prolonged air-drying (5 seconds). 8-9 <i>(Surfaces with developmental defects such as enamel hypomineralization (including fluorosis), tooth wear (attrition, abrasion and erosion), and extrinsic or intrinsic stains will be recorded as sound).</i>
	Initial stage caries (ICDAS™ codes 1 and 2)	 First or distinct visual changes in enamel seen as a carious opacity or visible discoloration (white spot lesion and/or brown carious discoloration) not consistent with clinical appearance of sound enamel (ICDAS™ code 1 or 2) and which show no evidence of surface breakdown or underlying dentin shadowing.
	Moderate stage caries (ICDAS™ codes 3 and 4)	 A white or brown spot lesion with Localized enamel breakdown, without visible dentin exposure (ICDAS™ code 3), or an Underlying dentin shadow (ICDAS™ code 4), which obviously originated on the surface being evaluated. <i>(To confirm enamel breakdown, a WHO/CPI/PSR ball-end probe can be used gently across the tooth area - a limited discontinuity is detected if the ball drops into the enamel micro-cavity/discontinuity).</i>
	Extensive stage caries (ICDAS™ codes 5 and 6)	 A distinct cavity in opaque or discolored enamel with visible dentin (ICDAS™ code 5 or 6). <i>(A WHO/CPI/PSR probe can confirm the cavity extends into dentin).</i>

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Australian Dental Therapeutic Guidelines

Rubber Dam Clamps – Easy Table

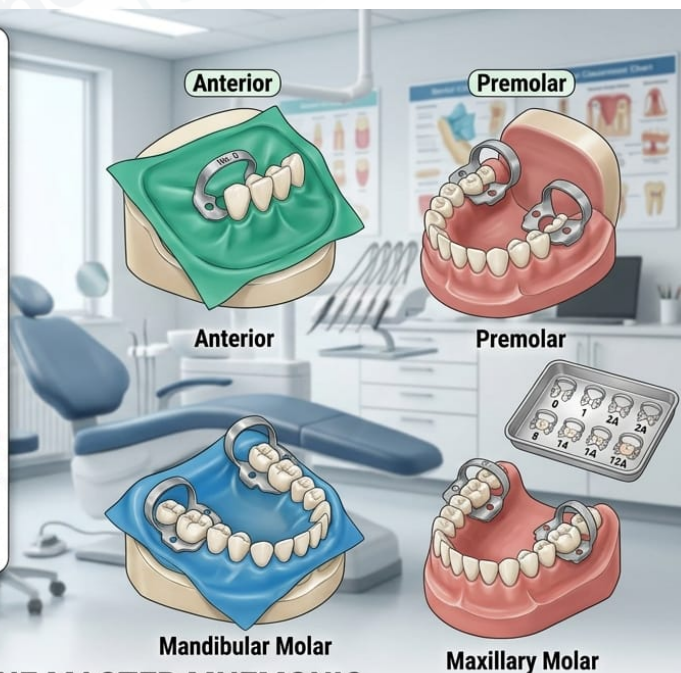
1. ANTERIOR TEETH 	Teeth: Incisors & Canines Common Clamp: No. 0 Mnemonic: "0 = Only Anterior"
2. PREMOLARS 	Teeth: Premolars Common Clamps: No. 1, No. 2, No. 2A, No. 212 Mnemonic: "P = 212 Party (Premolars)" or "2-2-12 for Premolars"
3. MANDIBULAR MOLARS 	Teeth: Lower molars Common Clamps: No. 8, No. 14, No. 14A Mnemonic: "8 & 14 = Lower 4 teeth (molars)"
4. MAXILLARY MOLARS 	Teeth: Upper molars Common Clamps: No. 12A, No. 13A, No. 14A Mnemonic: "12-13-14 = Upper Molars Trio"

SUPER QUICK REVISION TABLE

Anterior	0	"Only Anterior"
Premolars	1, 2, 2A, 212	"2-2-12 Premolars"
Mandibular molars	8, 14, 14A	"8 & 14 Lower molars"
Maxillary molars	12A, 13A, 14A	"12-13-14 Upper molars"

ONE-LINE MASTER MNEMONIC

 "0 for Anterior, 2-2-12 for Premolars, 8-14 for Lower Molars, 12-13-14 for Upper Molars"



GENERAL PRINCIPLES OF ENDODONTIC ACCESS (OCCT)

1. OUTLINE FORM



Definition: Shape & Position based on internal pulp anatomy.

Purpose: Locate canals, remove roof, provide straight-line access. Avoid missing canals.

Example: Maxillary central incisor → Triangular.

Memory Trick: Outline = Shape of pulp projected.



2. CONVENIENCE FORM



Definition: Modification of outline for easy instrumentation.

Purpose: Unobstructed access, direct apex access, easier file use, obturation space.

Clinical Consequence If Inadequate:

File separation, ledge formation, perforation.

3. CARIES REMOVAL



Definition: Remove all carious dentin & restorations first.

Why Important:

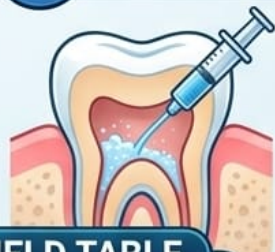
Reduce bacteria, assess restorability, sound structure for isolation.

Memory Trick:

No clean tooth = No clean root canal.



4. TOILET OF THE CAVITY



Definition: Complete cleaning of the pulp chamber after preparation.

Purpose: Prevent blockage, reduce bacterial load, visibility, prevent staining.

Clinical Consequence If Neglected:

Canal blockage, infection.

ADC HIGH-YIELD TABLE

	OCCT	EASY MNEMONIC
O	Outline Form	Outline Form
C	Convenience	Convenience Form
C	Caries	Inteoporates Form
T	Toilet of Cavity	Toilet of Cavity

O-C-C-T =

Outline Form, Convenience Form, Caries Removal, Toilet of Cavity.

COMPLETE DENTURE

► Most important support area in maxilla → **Horizontal hard palate.**

► Primary stress-bearing area of maxilla → **Posterolateral slopes of hard palate.**

► Primary stress-bearing area of mandible → **Buccal shelf.**

► Relief area in mandible → **Mental foramen.**

► Relief area in maxilla → **Incisive papilla and mid-palatine raphe.**

► Posterior palatal seal compensates for **polymerization shrinkage.**

► House classification describes **soft palate form.**

► Class I soft palate is **most favorable** for denture retention.

► Class III soft palate is **least favorable.**

MAXILLA

Most important support area: **Horizontal hard palate**

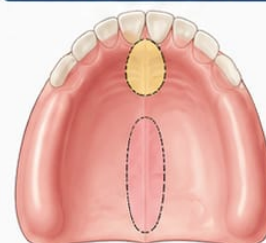
Primary stress-bearing area: **Posterolateral slopes of hard palate**

MANDIBLE

Primary stress-bearing area: **Buccal shelf**

Relief area: **Mental foramen**

RELIEF AREAS IN MAXILLA



- Incisive papilla
- Mid-palatine raphe

POSTERIOR PALATAL SEAL



Compensates for polymerization shrinkage

HOUSE CLASSIFICATION (Soft Palate Form)

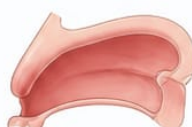
CLASS I

Most favorable for denture retention



CLASS II

Moderately favorable



CLASS III

Least favorable for denture retention



IMPRESSION MATERIALS

ZINC OXIDE EUGENOL (ZOE)



- Rigid set
- Mucostatic
- Wash impression

ALGINATE



- Irreversible hydrocolloid
- Economical
- Preliminary impression

COMPOUND



- Thermoplastic
- Reusable
- Border molding

POLYSULFIDE (RUBBER BASE)



- Elastomeric
- Good detail
- Final impression

SILICONE (ADDITION)



- Elastomeric
- Excellent detail
- Dimensional stability

SILICONE (CONDENSATION)



- Elastomeric
- Economical
- Good detail

ARTICULATORS

DEFINITION: An articulator is an instrument that simulates the movements of the mandible in relation to the maxilla.

1 MEAN VALUE ARTICULATOR

Non-adjustable articulator.

Uses average values



USAGE

- For routine denture cases
- When patient specific records are not available
- Simple and economical

2 SEMI-ADJUSTABLE ARTICULATOR

Most commonly used in clinics.

Examples:
Hanau, Denar, Whip-Mix



USAGE

- Most widely used for RPD and complete dentures
- Allows adjustment of condylar inclination and Bennett angle

3 FACEBOW TRANSFER

Relates maxillary cast to hinge axis / condylar axis



USAGE

- Transfers maxillary cast position in relation to the hinge axis
- Helps in accurate occlusion and jaw relation

4 ARCON ARTICULATOR

Resembles natural TMJ anatomy.

Condyle is on lower member



USAGE

- Closest resemblance to natural TMJ
- Used in complex cases and research



IMPORTANCE: Articulators help in mounting casts, recording jaw relations, setting teeth and achieving balanced occlusion.

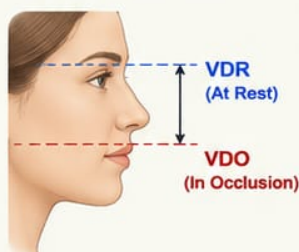


VERTICAL DIMENSION

DEFINITION: The vertical dimension is the measurement of the lower third of the face when the mandible is in a given position.

5 VDR > VDO

Vertical dimension at rest is always greater than VDO.

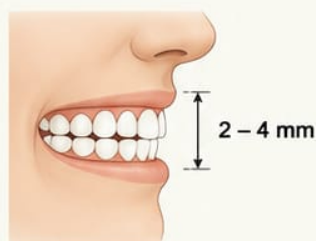


KEY POINT

There is a freeway space between rest and occlusion.

6 FREEWAY SPACE / INTEROCCLUSAL REST SPACE

Normally 2 – 4 mm



KEY POINT

Physiologic rest space between VDR and VDO.

7 EXCESSIVE VDO

Causes:

- Clicking of teeth
- Muscle fatigue
- TMJ pain
- Difficulty in function
- Overstretched muscles



KEY POINT

Too much vertical dimension leads to pain and dysfunction.

8 REDUCED VDO

Causes:

- Angular cheilitis
- Overclosure
- Facial aging (wrinkles)
- Decreased lower facial height
- Muscle strain



KEY POINT

Too little vertical dimension leads to esthetic and functional problems.



SUMMARY: Correct selection of articulator and proper vertical dimension are essential for successful prosthodontic treatment.



ANTICOAGULANT & ANTITHROMBOTIC DRUGS



Antithrombotic drugs prevent the formation and growth of thrombi (clots) in blood vessels. They are used to prevent stroke, myocardial infarction, venous thromboembolism (DVT, PE), and to maintain patency of vascular grafts and catheters.

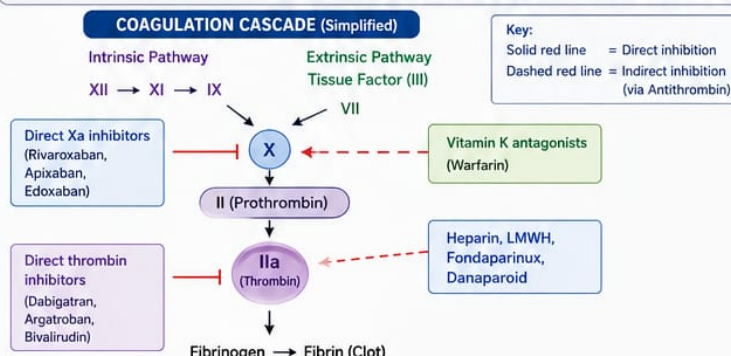
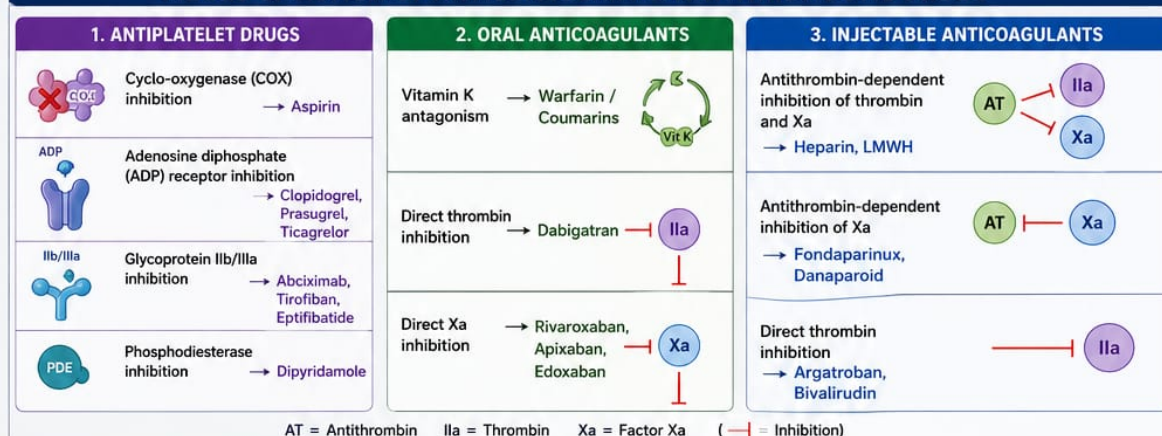


DRUG CLASS / EXAMPLES	INDICATIONS	CONTRAINDICATIONS	ADVERSE EFFECTS
ANTIPLATELET DRUGS - Aspirin - Clopidogrel - Prasugrel - Ticagrelor - Abciximab, Tirofiban, Eptifibatide - Dipyridamole	- Prevention of arterial thrombosis (MI, stroke, TIA) - Acute coronary syndromes - Post-PCI (with stent) - Peripheral arterial disease - Secondary prevention after ischemic stroke or TIA	- Active bleeding - Bleeding diathesis - Severe thrombocytopenia - Aspirin: hypersensitivity, active peptic ulcer - Intracranial hemorrhage - Recent major surgery	- Bleeding (GI, intracranial) - GI irritation / ulceration (aspirin) - Dyspepsia, nausea - Thrombocytopenia - Hypersensitivity reactions (Abciximab may cause thrombocytopenia)
ORAL ANTICOAGULANTS - Warfarin (Coumarins) - Dabigatran - Rivaroxaban - Apixaban - Edoxaban	- Prevention & treatment of venous thromboembolism (DVT, PE) - Prevention of stroke in non-valvular atrial fibrillation - Treatment of VTE - Mechanical heart valves (Warfarin)	- Active bleeding - Severe liver disease - Pregnancy - Mechanical heart valves (for DOACs) - Severe renal impairment (for some DOACs)	- Bleeding - Anemia - Dyspepsia (Dabigatran) - Hepatic toxicity (rare) - Skin rash - Teratogenicity (Warfarin) - Warfarin: skin necrosis (rare)
INJECTABLE ANTICOAGULANTS - Heparin (UFH) - Low Molecular Weight Heparin (LMWH) - Fondaparinux - Danaparoid - Argatroban - Bivalirudin	- Treatment & prophylaxis of DVT, PE - Acute coronary syndromes - Perioperative anticoagulation - Prevention of clotting in extracorporeal circuits (CPB, dialysis) - HIT (Argatroban, Bivalirudin)	- Active bleeding - Heparin-induced thrombocytopenia (for heparin, LMWH) - Severe thrombocytopenia - Uncontrolled hypertension - Recent CNS hemorrhage	- Bleeding - Heparin: HIT, osteoporosis - Thrombocytopenia - Injection site reactions - Hyperkalemia (Heparin) - Elevated liver enzymes (Argatroban)



Key Point: All antithrombotic drugs increase the risk of bleeding. Assess bleeding risk before initiation and monitor appropriately.

MODES OF ACTION OF ANTICOAGULANT & ANTITHROMBOTIC DRUGS



GENERAL ADVERSE EFFECTS (ALL ANTITHROMBOTICS)

- Bleeding – the most common and serious adverse effect
- Anemia
- Drug interactions (important)
- Monitor renal & hepatic function
- Patient education: report any signs of bleeding (e.g., black stools, blood in urine, prolonged bleeding from cuts)

Always individualize therapy based on indication, patient risk factors, and bleeding risk.



PHONETICS IN COMPLETE DENTURE AND THEIR IMPORTANCE



SOUND	WHAT IT CHECKS	CLINICAL IMPORTANCE	ILLUSTRATION / EXAMPLE
S "S"	Checks closest speaking space (1–2 mm) and vertical dimension (VDO)	• Clicking during S sound → VDO too high • Whistling during S sound → excessive space / incorrect anterior tooth position	
F "F"	Checks maxillary incisor length and position	• F distorted → upper incisors too long, too short, or wrongly positioned	
V "V"	Checks maxillary incisors touching wet-dry line of lower lip	• F/V distorted → upper incisors too long, too short, or wrongly positioned	
M "M"	Determines rest position of mandible (VDR) Used to calculate freeway space		Say "MMM" → Mandible in rest position (VDR) → Freeway space (2–4 mm)
P "P"	Checks lip support and anterior tooth position		
B "B"	Checks lip competence and tooth position		
P, B, M "P", "B", "M"	Assess labial flange thickness and lip support		
T "T"	Checks tongue contact with anterior palate		
D "D"	Evaluates maxillary anterior tooth position		
N "N"	Assesses tongue-palate relationship		
T, D, N "T", "D", "N"	Evaluate lingual contour of denture		
K "K"	Checks posterior palatal contour		
G "G"	Checks posterior tongue space and palatal thickness		
K / G distorted	K/G distorted → posterior palate may be overextended or too thick		



Phonetics is a simple, effective and non-invasive method to evaluate and refine complete dentures for a natural, comfortable and functional outcome.



Clinical Reference Image

Australian Dental Therapeutic Guidelines

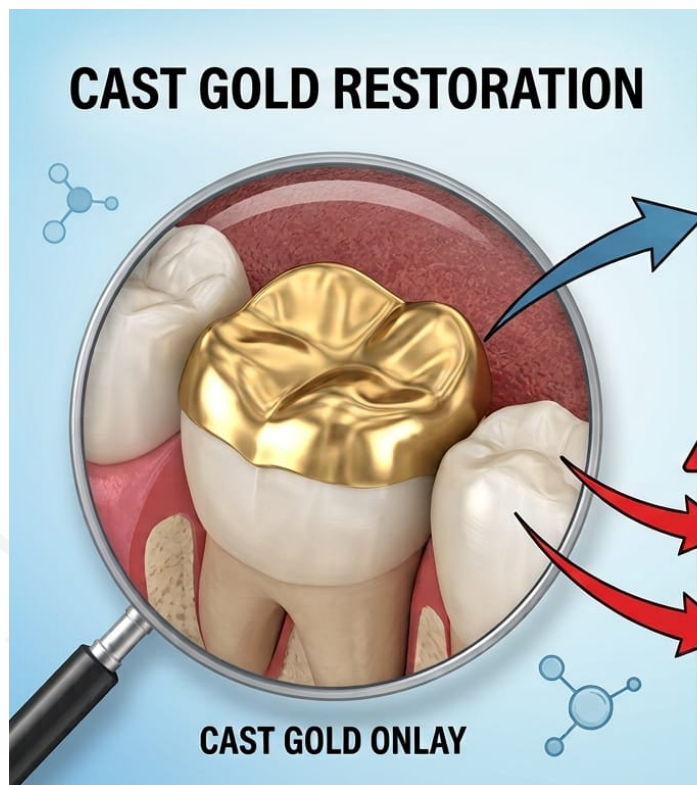
 DENTISTRY INSIGHT WELFARE ASSOCIATION DENTISTRY LINEAR POINTS High-Yield Diagnostic & Clinical Pearls					
	ONE-LINER	DISEASE / CONDITION	DIAGNOSTIC POINT	CLINICAL PEARL	
1	 Most common odontogenic cyst	Radicular cyst	• Unilocular radiolucency at the apex of a non-vital tooth	• Treat with root canal therapy (if vital) or enucleation (if non-vital)	
2	 Most common developmental odontogenic cyst	Dentigerous cyst	• Unilocular radiolucency associated with the crown of an unerupted tooth	• Enucleation (marsupialization in large lesions)	
3	 Most common odontogenic tumour	Odontoma	• Well-defined radiopaque mass (compound) or irregular radiopaque mass with a radiolucent rim (complex)	• Enucleation; rare recurrence	
4	 Most common true odontogenic neoplasm	Ameloblastoma	• Multilocular "soap-bubble/honeycomb" radiolucency, often in posterior mandible	• Enucleation or resection depending on size; may recur	
5	 Soap-bubble radiolucency	Ameloblastoma	• Multilocular radiolucency with thin septa giving soap-bubble appearance	• Usually Ameloblastoma	
6	 Ground-glass appearance	Fibrous dysplasia	• Diffuse, hazy radiopacity resembling ground glass	• Suggests Fibrous dysplasia	
7	 Cotton-wool appearance	Paget disease	• Ill-defined mixed radiolucent-radiopaque areas giving cotton wool appearance	• Seen in Paget disease	
8	 Sunburst appearance	Osteosarcoma	• Radiating spicules of bone from periosteal surface	• Suggests Osteosarcoma	
9	 Onion-skin appearance	Ewing sarcoma	• Concentric layers of periosteal new bone	• Suggests Ewing sarcoma	
10	 Driven-snow appearance	Calcifying epithelial odontogenic tumour (Pindborg tumour)	• Fine, diffuse radiopacities resembling driven snow	• Seen in Calcifying epithelial odontogenic tumour (Pindborg tumour)	
11	 Ghost cells	Calcifying odontogenic cyst (Gorlin cyst)	• Eosinophilic epithelial cells with absent nuclei	• Characteristic of Calcifying odontogenic cyst (Gorlin cyst)	
12	 Cherry-red gingiva	Carbon monoxide poisoning	• Bright red gingiva	• Seen in Carbon monoxide poisoning	
13	 Punched-out interdental papillae	Acute necrotizing ulcerative gingivitis (ANUG)	• Necrosis of interdental papillae with ulceration	• Seen in ANUG	
14	 Wickham striae	Oral lichen planus	• White, lacy, reticular lines on oral mucosa	• Pathognomonic of Oral lichen planus	
15	 Positive Nikolsky sign	Pemphigus vulgaris	• Epithelium separates on gentle lateral pressure	• Seen in Pemphigus vulgaris	
16	 Target lesions	Erythema multiforme	• Lesions with concentric rings: central dusky area, pale zone, erythematous border	• Seen in Erythema multiforme	
17	 White plaques that wipe off	Pseudomembranous candidiasis	• White lesions that can be scraped off leaving an erythematous base	• Suggests Pseudomembranous candidiasis	
18	 White lesion that cannot be scraped off	Leukoplakia	• White patch persists, non-removable by scraping	• Suggests Leukoplakia	
19	 Red oral lesion with high malignant potential	Erythroplakia	• Velvety red patch or plaque	• Suggests Erythroplakia	
20	 Most common oral malignancy	Squamous cell carcinoma	• Ulceroproliferative growth, indurated margins; may be exophytic or ulcerative	• Squamous cell carcinoma	
21	 Common high-risk site for oral SCC	Lateral border of tongue	• Most frequent site for oral SCC	• Lateral border of tongue	
22	Oral Kaposi sarcoma commonly affects	Hard palate	• Purple-red macules/nodules	• Hard palate	
 KEY TAKEAWAY Recognize classic patterns, appearances, and presentations to make accurate and quick diagnoses. 					

ADC TG 4: MANAGEMENT OF ACUTE ALLERGIC REACTIONS

1. ACUTE URTICARIA & ANGIOEDEMA (LOCALIZED)	2. ANAPHYLAXIS (SYSTEMIC COLLAPSE)	3. ANAPHYLAXIS EMERGENCY PROTOCOL					
CLINICAL FEATURES  Itchy, raised red wheals Non-pitting subcutaneous swelling Skin reaction only MANAGEMENT Stop treatment/remove trigger → Assess Airway (No obstruction/wheeze) Position: Comfortably seated Administer oral non-sedating antihistamine (CETIRIZINE 10mg) Observe (30 mins) → Observe (30 mins) → Refer to GP	CLINICAL FEATURES  A (Airway): Throat tightness, hoarse voice B (Breathing): Sudden bronchospasm, wheeze C (Circulation): Hypotension, pale, weak pulse Skin: Rapidly spreading hives LIFE-THREATENING	CALL 000 IMMEDIATELY Position: LAY FLAT, RAISE LEGS Administer IM ADRENALINE (Mid-Thigh) ADRENALINE 1:1,000 IM (anterolateral thigh) <table border="1"> <tr> <td>ADULT DOSE:</td> <td>0.5 mg (0.5 mL)</td> <td rowspan="2">Wait 5 minutes</td> </tr> <tr> <td>CHILD DOSE:</td> <td>Based on weight (e.g., <20kg: 0.15 mg, >20kg: 0.3 mg)</td> </tr> </table> IF NO IMPROVEMENT Repeat Adrenaline dose every 5 minutes → Monitor ABC/Airway support ADDITIONAL MEASURES High-flow Oxygen (15L/min) → Transfer to hospital for biphasic observation NO Sedating Antihistamines first-line	ADULT DOSE:	0.5 mg (0.5 mL)	Wait 5 minutes	CHILD DOSE:	Based on weight (e.g., <20kg: 0.15 mg, >20kg: 0.3 mg)
ADULT DOSE:	0.5 mg (0.5 mL)	Wait 5 minutes					
CHILD DOSE:	Based on weight (e.g., <20kg: 0.15 mg, >20kg: 0.3 mg)						



CAST GOLD RESTORATION



CAST GOLD ONLY

CAST GOLD ADVANTAGES (STANDARD)

- A range of alloys available
- Highly accurate fit available
- Reinstates full coronal anatomy
- Wear factor similar to tooth structure
- Longevity may justify cost

! CAST GOLD DISADVANTAGES (STANDARD)

- Multi-stage production allows for error
- High skill required at all stages
- Relatively high cost initially
- Long-term dissolution of the luting agent possible
- Aesthetics a matter of choice

Clinical Reference Image

Australian Dental Therapeutic Guidelines

BACKGROUND

- * "CANKER SORES"
- * SMALL, PAINFUL ULCERS
- * FORM on **NON-KERATINIZED ORAL MUCOSA**
 - ~ INNER LIPS, CHEEKS, SOFT PALATE, FLOOR of MOUTH, GINGIVA & TONGUE
- * AFFECTS 20% of POPULATION

SIGNS & SYMPTOMS

- * PINPOINT PAPULE, GRADUALLY EXPANDS
- * WELL CIRCUMSCRIBED
- * ROUND / OVAL, YELLOW GRAY PSEUDOMEMBRANOUS CENTER, INFLAMMATORY RED HALO
- * HEALS within 7-14 DAYS

TYPES

MINOR
(< 1 CM)

MAJOR
($> 2-3$ CM)

HERPETIFORM
(MULTIPLE)



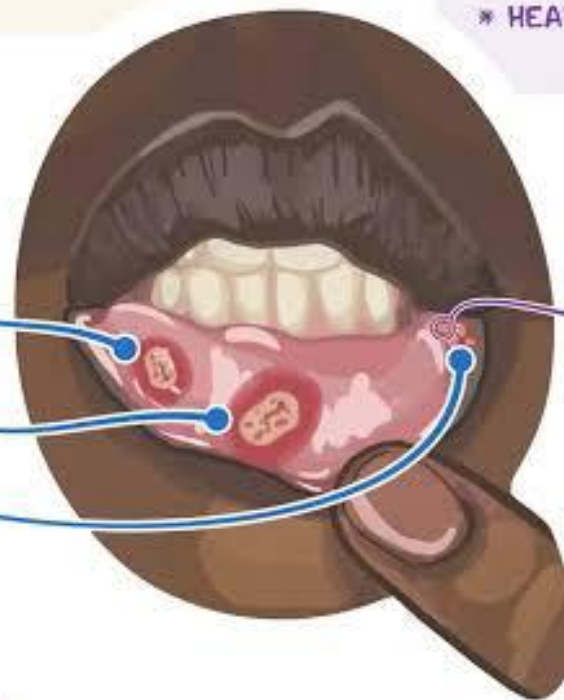
TREATMENT

- * MANAGE STRESS
- * HEALTHY DIET
- * ORAL HYGIENE



CAUSES

- * WEAKENED IMMUNE SYSTEM
- * CERTAIN FOODS
- * $\uparrow\uparrow$ STRESS
- * NUTRITIONAL DEFICIENCY





CLINICAL FEATURES, CLASSIFICATION AND TREATMENT OF PERIODONTAL DISEASE

1. CLINICAL EXAMINATION

The elements of a clinical examination of the periodontal tissues include assessment of gingival appearance and periodontal attachment level (screening assessment or complete periodontal charting). A comprehensive clinical examination should also assess the local factors that contribute to biofilm accumulation, interfere with adequate oral hygiene or contribute to damage to the periodontal tissues.

VISUAL INSPECTION

Gingival inflammation, position and contour, redness, swelling, enlargement, recession or muco-gingival problems.
Evidence of pathological conditions (ulcerations or other mucosal conditions).
Oral hygiene (biofilm deposits – site/volume) related gingival inflammation.

SCREENING ASSESSMENT - Basic Periodontal Examination (BPE)

Screen all new clients (children and adults) presenting for general course of care. BPE recorded at every recall examination with Codes 0, 1 or 2 on previous exams. Conduct a complete periodontal charting and radiographic assessment in:
• All cases of Code 3 (for all teeth in sextant) and Code 4 (entire Dentition)
• All cases of complex care – oral rehabilitation with fixed or removable prosthesis

COMPLETE PERIODONTAL CHARTING - provides evidence of:

Clinical Attachment Loss (CAL) - Periodontal probing depth (6-point)
- Recession (6-point)
- Furcation involvement (Grade 1, 2, 3)
- Mobility (Grade 1, 2, 3)
Inflammation (including contributing factors) - Bleeding on Probing (BOP)
- Plaque Score/Index

RADIOGRAPHS

Assess crestal bone level, plaque-retentive factors (calculus, restorative margins) tooth/root morphology and anomalies, and evidence of pathology (dental, periodontal, periapical).

2. CLINICAL FEATURES

1 GINGIVAL CHANGES

- Redness
- Swelling
- Bleeding on probing
- Tenderness

2 POCKET FORMATION

- Probing depth > 3 mm
- Gingival pocketing
- Attachment loss

3 ALVEOLAR BONE LOSS

- Radiographic bone loss
- Increased tooth mobility
- Furcation involvement

4 PLAQUE & CALCULUS DEPOSITS

- Visible deposits
- Halitosis
- Staining

5 FUNCTIONAL EFFECTS

- Tooth mobility
- Migration
- Occlusal changes
- Difficulty in chewing

Note: a screening assessment (BPE) should not be used for diagnosis nor replace a complete periodontal examination. It is inadequate to monitor the response to periodontal therapy or to monitor the periodontal condition for clients in Maintenance/ Supportive Periodontal Therapy.

Note: The overall Clinical Assessment may identify features that could contribute to the initiation/progression of disease or will influence the provision and outcome of treatment. These features may include:

- Tooth position and malocclusion - (rotation, tipping, supra-eruption)
- Food impaction
- Inadequate restoration
- Traumatic occlusal contact or function
- Fixed and removable prostheses or orthodontic appliances.

Additional Tests

Additional clinical investigations may be required to assist in the formulation of a definitive diagnosis and treatment plan; including tests for caries, endodontic and/or periodontal status and restorability.

3. CLASSIFICATION (DISEASE PROGRESSION)



4. TREATMENT (MANAGEMENT STRATEGIES)

NON-SURGICAL (BASIC THERAPY)

- Oral Hygiene Instruction**
Patient education and motivation
- Scaling and Root Planing**
Removal of plaque and calculus
- Adjunctive Aids**
Mouthrinses, local antimicrobials (if indicated)
- Review and Re-evaluation**
Assess response and modify plan

SURGICAL / ADVANCED THERAPY

- Flap Surgery**
Access for deep cleaning and defect correction
- Bone Regeneration**
Grafting procedures to restore lost bone
- Guided Tissue Regeneration**
Use of membranes to support healing
- Tooth Replacement**
Implants, bridges or removable prostheses when needed

5. MAINTENANCE AND PREVENTION

- Maintain excellent oral hygiene
- Follow a balanced diet
- Avoid tobacco use
- Regular dental check-ups
- Early diagnosis saves teeth

Healthy gums, strong support – for a lifetime of confident smiles.

* Complete Periodontal Charting: provides evidence of Clinical Attachment Loss (CAL), probing depth (6-point), recession (6-point), recession/furcation, furcation involvement, supra-eruption, calculus, staining, mobility (Grade 1, 2, 3), bleeding on probing (BOP) and plaque score/index.